

Causal Inverse Reinforcement Learning for Robust Reward Recovery

Govind Arun Nampoothiri



Master of Science
Data Science
School of Informatics
University of Edinburgh
2025

Abstract

Reward functions are central to reinforcement learning but are rarely specified directly in practice. Inverse reinforcement learning (IRL) seeks to recover rewards from expert demonstrations, yet existing methods are brittle under latent confounding, i.e., when hidden factors shape both state visitation and actions. This dissertation investigates whether causal structure can improve robust reward recovery. We develop Causal-AIRL, a minimal extension of Adversarial IRL that conditions the discriminator on a latent variable Z and penalises per- (s, a) reward variance across Z to encourage invariance. Our contributions are fourfold: (i) a causally informed adversarial IRL objective; (ii) a controlled GridWorld benchmark with simulated confounding; (iii) reward-level invariance metrics and overlays; and (iv) a figure-first evaluation pipeline spanning clean sweeps, confounded experts, held-out generalisation, and scaling.

Empirically, Causal-AIRL improves cross- Z policy agreement under confounding with non-overlapping 95% confidence intervals relative to AIRL (Fig 6.1), while matching AIRL on clean reward fidelity across demonstrations, slip, reward type, and discount factors (Fig 6.5 - Fig 6.7). A matched-scenario variance slice indicates lower across- Z reward variance for Causal-AIRL (Fig 6.2), and training diagnostics show non-degenerate learning with a well-behaved latent and earlier policy-loss stabilisation (Fig 6.9). Held-out generalisation retains these transfer gains (Fig 6.4). Compute overhead is modest: scaling curves place Causal-AIRL near AIRL on the accuracy–time frontier (Fig 6.8). The scope is deliberately restricted to tabular GridWorlds with known dynamics and simulated latent confounders; results should be read as controlled evidence that causal invariance aids transfer without degrading clean-regime fidelity. The implication is that invariance-regularised reward learning offers a principled route to more robust reward recovery in settings where demonstrations plausibly encode hidden variation.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Govind Arun Nampoothiri)

Acknowledgements

My sincere thanks to my supervisor, Prof. Michael Herrmann, for the chance to pursue this work and for his thoughtful guidance, clear-eyed feedback, and steady encouragement. I'm also grateful to the Informatics staff and my cohort for practical help and many useful conversations. Above all, thank you to my family and friends for their patience and good humour through late nights and early mornings. Any errors are mine alone.

Table of Contents

1	Introduction	1
2	Related Work	4
2.1	Classical Inverse Reinforcement Learning	4
2.2	Deep and Adversarial IRL	5
2.3	Causality in Imitation and Reinforcement Learning	5
2.4	Human Alignment and Preference Learning	6
2.5	Synthesis and Timeliness	6
3	Background & Preliminaries	8
3.1	Markov Decision Processes	8
3.2	Inverse Reinforcement Learning	9
3.3	Adversarial Inverse Reinforcement Learning	9
3.4	Causal Inference and Structural Models	10
3.5	Notation Summary	11
4	Methods	13
4.1	Classical IRL Baselines	13
4.1.1	Ng and Russell (Projection IRL)	13
4.1.2	Maximum Entropy IRL (MaxEnt)	14
4.2	Adversarial Inverse Reinforcement Learning	14
4.2.1	Objective and Reward	15
4.2.2	Decomposition and Transfer	15
4.3	Causal-AIRL	16
4.3.1	Latent Conditioning	16
4.3.2	Invariance-Regularised Objective	16
4.4	Implementation Notes and Guardrails	17

5	Experiments	19
5.1	Environment and controllable factors	19
5.2	Evaluation questions and families	20
5.3	Metrics and aggregation	21
5.4	Setups in detail	21
5.5	Method-specific instantiation	23
5.6	Sanity checks and failure guards	23
5.7	Assumptions, scope, and reproducibility	23
6	Results	25
6.1	Classical IRL (Ng–Russell, MaxEnt)	25
6.2	AIRL (reference comparator)	26
6.3	Causal-AIRL	26
6.4	Training diagnostics (context)	30
7	Discussion	36
8	Conclusions	39
	Bibliography	41
A	Reproducibility Details	44
B	Hyper-parameters	47
C	Notation and Symbols	48
D	Hardware, Software, and Scaling	49
E	Aggregated Tables & CI Summaries	51
E.1	Cross-Z Transfer: Policy Agreement by Method and Demonstrations .	51
E.2	Held-out Generalisation: Confounded + Spatial Masks	52
E.3	Scaling Curves: Accuracy, Time, and Sample Use	54
E.4	Method-wise Summary (All Runs)	56
E.5	Reward Type $\times$ Method Summary	57

Chapter 1

Introduction

Reinforcement learning (RL) frames sequential decision-making as interaction with an environment in order to maximise long-run utility. An agent observes a state, selects an action, receives a scalar reward, and the environment transitions to a new state; the goal is to learn a policy that achieves high expected return. This formalism has proved useful across domains where objectives are natural but hard to encode precisely: autonomous driving (comfort, safety, progress), assistive robotics (user intent and ergonomics), recommendation and tutoring systems (engagement without harm), and logistics or healthcare scheduling (fairness and timeliness). In all such settings the practical bottleneck is not optimisation, but specification: what the agent *ought* to do is hard to write down as a single reward.

Inverse reinforcement learning (IRL) tackles this specification problem by learning a reward from expert demonstrations. Rather than prescribing the objective directly, we ask which reward would render the expert’s behaviour near-optimal and then seek a policy that is optimal for that reward. This differs from behavioural cloning, which directly imitates actions; cloning can be effective on well-covered states but is brittle under distribution shift (see DAgger for covariate-shift correction [23] and broader overviews of imitation learning practice [17]). Consider a sparse-goal grid where an expert occasionally detours to avoid narrow corridors: a cloning policy may copy the detour and fail when the local geometry changes, whereas a reward that captures the underlying goal and preferences can guide sensible behaviour in new states.

IRL, however, is fundamentally underdetermined. In a Markov decision process $(\mathcal{S}, \mathcal{A}, T, R, \gamma)$, multiple reward functions can induce the same optimal policy, and potential-based shaping leaves optimal behaviour unchanged. Even when a reward is identifiable up to certain transformations, limited demonstration coverage, noise,

or systematic suboptimality further blur the picture. Most crucially for real data, demonstrations often reflect latent factors—operator style, risk tolerance, time pressure, unobserved constraints—that influence both the states visited and the actions taken. These *confounders* entangle intent with circumstance, so a learner that treats demonstrations as arising from a single, stationary process may attribute context-specific artefacts to the reward itself.

This dissertation explores whether making such latent structure explicit improves the robustness of reward recovery. We compare classical IRL (Ng–Russell; Maximum Entropy IRL), Adversarial IRL (AIRL), and Causal-AIRL, an AIRL variant that conditions on a latent Z and penalises the per- (s, a) reward variance across Z to encourage invariance. The empirical setting is deliberately controlled: fully observed, tabular GridWorlds with known dynamics and simulated Z . Within this scope we probe robustness by varying reward density, transition slip, demonstration count, and discount factor, and by constructing confounded experts and held-out regions to test generalisation. Reward fidelity is measured primarily by rank correlation (Spearman), with value correlation and policy agreement as secondary checks; invariance is summarised by the across- Z reward variance. Notation is standard and consolidated in Chapter 3.

Methodologically, Causal-AIRL augments AIRL’s discriminator–generator game with latent conditioning and a lightweight invariance regulariser. Conditioning exposes the discriminator to Z -dependent variation, while the penalty discourages reward assignments that fluctuate across Z at the same (s, a) while preserving AIRL’s potential-based shaping. The design is intentionally minimal—no changes to the overall adversarial structure—so differences can be attributed to the treatment of latent variation rather than to architectural confounds. Empirically, we conduct a controlled comparison across clean sensitivity sweeps, confounding, held-out generalisation, and compute–accuracy scaling, all within the same evaluation rubric.

This study is timely for three reasons. First, demonstrations increasingly come from heterogeneous operators and contexts, and deployment often requires crossing latent domains. Second, deep IRL methods have matured to the point where reward recovery is feasible beyond tiny tabular settings, yet they remain sensitive to hidden variables that bias occupancy and feedback. Third, ideas from causal modelling and invariance have clarified when structure aids generalisation—e.g., invariant causal prediction [19] and invariant risk minimisation [2]—suggesting a principled route to robustness without discarding the advantages of adversarial IRL. Within our controlled scope we show that a causally informed objective can improve cross-context transfer

while maintaining parity in clean regimes and incurring only modest additional compute.

We preview the arc and where the evidence appears: Chapter 2 positions the work in IRL, imitation, and causal invariance; Chapter 3 sets notation and causal assumptions; Chapter 4 instantiates the compared methods; Chapter 5 details protocols and metrics; Chapter 6 reports outcomes—cross- Z transfer (Fig. 6.1) and held-out generalisation (Fig. 6.4), clean-regime fidelity (Figs. 6.5–6.7), and accuracy–time plus diagnostics (Figs. 6.8–6.9); Chapter 7 synthesises implications and limits; Chapter 8 concludes. Along the way, our contribution is to frame robust reward recovery through a causal lens and make it concrete: we embed a latent Z into AIRL and regularise reward variability to promote invariance; we construct a controlled confounded benchmark; and we align evaluation metrics and figure-first reporting with the goal of transferring policy-useful rewards under latent shift (reproducibility and configuration details in A; compute notes in D).

Chapter 2

Related Work

This chapter situates the thesis within inverse reinforcement learning (IRL) and neighbouring areas. We begin with classical IRL in tabular settings, proceed to deep and adversarial formulations that learn neural rewards, and then consider causal perspectives in imitation and IRL aimed at robustness under hidden variables and distribution shift. A brief connection to alignment via human feedback closes the loop. The emphasis is on methods that produce explicit rewards suitable for transfer, as our evaluation is entirely in tabular GridWorld with known dynamics (Chs. 4–5). For general background on RL and causal modelling see Sutton and Barto [26] and Pearl [18], and for imitation learning foundations and covariate-shift corrections see Ross et al. [23] and Osa et al. [17].

2.1 Classical Inverse Reinforcement Learning

Ng and Russell [16] cast IRL as searching for a reward under which the expert is optimal, proposing a projection method that enforces optimality via (feature) visitation constraints. Maximum Margin Planning [22] distinguishes expert trajectories from alternatives by a large margin, while Maximum Entropy IRL (MaxEnt) [30] selects, among all distributions matching expert feature expectations, the one of maximum entropy, yielding a probabilistic, noise-tolerant estimator. Apprenticeship learning via occupancy matching [1] provides the occupancy-based perspective adopted by many later methods. Maximum Causal Entropy IRL (MCE) [29] strengthens the formulation by imposing causal conditioning, preventing non-causal dependence on future stochasticity. A complementary strand frames apprenticeship learning game-theoretically, linking margin-style guarantees with occupancy matching [27]. These approaches de-

liver interpretable, explicit rewards and remain strong baselines when dynamics are known and the state–action space is modest. Their limits are equally well understood: they typically assume fully observed, largely unconfounded demonstrations, and they struggle when features and dynamics must be learned. These constraints motivated deep formulations that learn reward representations directly from data.

2.2 Deep and Adversarial IRL

Deep IRL replaces hand-crafted features with neural reward models. Maximum Entropy Deep IRL [28] learns dense spatial reward maps from high-dimensional inputs, showing that MaxEnt-style objectives can be paired with convolutional encoders. Guided Cost Learning [6] integrates MaxEnt IRL with on-policy sampling in continuous control, demonstrating that neural rewards can be trained effectively with policy optimisation. As a related bridge between supervised learning and RL, guided policy search trains policies via supervised learning on trajectories produced by trajectory optimisers [15]. Adversarial IRL (AIRL) [7] frames IRL as a GAN-like (Generative Adversarial Network) game between a policy and a discriminator and introduces a reward decomposition that, under state-only, stationary rewards, supports dynamics-invariant transfer. Beyond AIRL, distribution-matching approaches (the DICE family) optimise divergence objectives to align expert and learner occupancy and typically produce a policy rather than an explicit reward [12, 9]. In practice, adversarial methods often presume full observability and fairly homogeneous data; when demonstrations reflect unobserved variation, the learned reward can inherit spurious correlations that degrade generalisation [5, 24]. This brittleness under hidden structure motivates bringing causal ideas to bear.

2.3 Causality in Imitation and Reinforcement Learning

Causal perspectives in imitation learning address causal confusion—policies that seize on non-causal cues and fail under shift [5]. Invariance principles [2, 25] propose learning mechanisms whose predictive relations remain stable across environments; ICIL (Invariant Causal Imitation Learning) [3] adapts this idea to imitation by encouraging policies invariant to environment-specific factors, building on invariant causal prediction [19]. Risk extrapolation (REx) complements IRM (Invariant Risk Minimization) by penalising variability of risks across environments to promote invariance [13].

Domain-adversarial training encourages domain-invariant features when environment labels are observed [8]; such labels mitigate—but do not eliminate—latent confounding. These approaches primarily target policies rather than rewards, improving robustness without returning an explicit, transferable objective. Closer to our setting, causal IRL seeks rewards that remain stable when unobserved variables influence both states and actions; Ruan et al. [24] incorporate causal constraints to improve reward fidelity under confounding. Within adversarial imitation, GAIL (Generative Adversarial Imitation Learning)-style discriminators conditioned on observed environment descriptors can mitigate confounding [11, 5], but they typically do not yield explicit rewards or handle latent confounders. What remains underexplored is an adversarial IRL formulation that integrates a latent confounder and explicitly regularises reward invariance while retaining AIRL’s desirable decomposition—this is the gap our method addresses.

2.4 Human Alignment and Preference Learning

A complementary line infers objectives from interaction rather than demonstrations alone. Cooperative IRL [10] models human–agent cooperation to reveal preferences; preference-based learning elicits pairwise trajectory judgements to train reward models [4]; and scalable reward-modelling frameworks [14] argue for learning evaluators at scale as a route to aligning complex systems. These approaches share our emphasis on accurate objectives but typically require supervision beyond passive demonstrations. Causal reward recovery from demonstrations is complementary: by filtering spurious correlations in passive data, explicit rewards can serve as stronger priors or starting points for interactive alignment.

2.5 Synthesis and Timeliness

Taken together, the literature supports a simple trajectory. Classical IRL provides clarity and explicit rewards but relies on known dynamics and largely unconfounded data. Deep and adversarial methods scale to richer observations and can, under structural assumptions, transfer across dynamics, yet they are sensitive to hidden variables that bias expert occupancy. Causal imitation and representation learning improve robustness, often by encouraging invariance at the policy level, but typically do not produce explicit reward functions or integrate with adversarial IRL objectives. This makes a causally informed, reward-explicit adversarial IRL particularly timely: demonstrations

increasingly come from heterogeneous operators and contexts, deployment requires crossing latent domains, and invariance principles have matured enough to inform objective design. We address this gap by embedding a latent confounder in AIRL and penalising the per- (s, a) reward variance across Z to encourage invariance, aiming to recover rewards that are both transferable and robust to confounding. The causal assumptions are made explicit in Chapter 3; the comparative evaluation in tabular Grid-World appears in Chapters 5–6. Read together, these strands motivate our focus on an adversarial IRL objective that retains explicit rewards while integrating a latent Z and an invariance penalty, with the goal of learning reward functions that remain policy-useful under confounding.

Table 2.1: Method landscape. Policy vs reward outputs; whether causal structure is used; and whether an explicit reward is produced.

Method	Output	Causal?	Explicit reward?	Note
Ng–Russell, MaxEnt IRL	Reward	×	✓	Tabular, known dynamics.
Guided Cost Learning	Reward	×	✓	Neural reward; continuous control.
Maximum Entropy Deep IRL	Reward	×	✓	CNN reward; dense spatial maps.
AIRL	Reward	×	✓	Decomposition; transfer under state-only, stationary rewards.
GAIL / Causal-GAIL	Policy	✓ [†]	×	Environment descriptors reduce confounding.
ValueDICE / DICE family	Policy	×	×	Distribution matching; typically no explicit reward.
ICIL / IRM-style	Policy	✓	×	Invariance across domains.
Causal-AIRL (our method)	Reward	✓	✓	Latent Z ; invariance-regularised.

[†] Uses observed environment descriptors; does not address latent Z .

Chapter 3

Background & Preliminaries

This chapter introduces the concepts used throughout the dissertation. We begin with Markov Decision Processes (MDPs) and the occupancy measures that link expert demonstrations to learning objectives; we then recap inverse reinforcement learning (IRL) in classical and adversarial forms; and we close with the causal tools that motivate our method. The goal is clarity rather than completeness: definitions come with brief intuition and notes on when standard assumptions can fail. Symbols are consolidated in Table 3.1. For foundational background see Sutton and Barto [26] and Puterman [21] on MDPs, and Pearl [18] and Peters–Janzing–Schölkopf [20] on causal models.

3.1 Markov Decision Processes

An MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \gamma)$ formalises sequential decision-making: at state $s \in \mathcal{S}$ ($\mathcal{S}$ is the set of all possible states) an agent chooses $a \in \mathcal{A}$ ($\mathcal{A}$ is the set of all possible actions), the environment transitions according to $T(s' | s, a)$, and a scalar reward $R(s, a)$ is delivered. The discount $\gamma \in [0, 1)$ weights near-term rewards more heavily. A (stochastic) policy $\pi_\phi(a | s)$ maps states to action probabilities and induces the return:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right], \quad (3.1)$$

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a \right]. \quad (3.2)$$

Beyond returns, it is useful to describe how a policy visits the state–action space. The (discounted) occupancy measure

$$\rho^\pi(s, a) = (1 - \gamma) \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \mathbb{1}_{\{(s_t, a_t) = (s, a)\}} \right] \quad (3.3)$$

captures the expected discounted frequency of encountering (s, a) , and normalisation gives $\sum_{s,a} \rho^\pi(s, a) = 1$. Occupancies connect demonstrations to learning objectives: matching expert occupancies is a principled route to imitating behaviour.

The Markov property assumes that T depends only on (s, a) , not on unobserved context. When hidden variables influence state visitation or action choices, the process seen by the learner can violate this assumption; effectively, the data arise from a mixture over latent contexts. We make such latent structure explicit in Section 3.4.

3.2 Inverse Reinforcement Learning

Given demonstrations $\mathcal{D}_E = \{\tau_i\}_{i=1}^N$, with each trajectory $\tau = (s_0, a_0, \dots, s_T)$ sampled from an unknown expert policy π_E , IRL seeks a reward R under which π_E is (approximately) optimal. Unlike behavioural cloning, which directly regresses $\pi_E(a|s)$, reward recovery targets the objective that explains the behaviour and thus supports optimisation in unseen states.

IRL is ill-posed: many distinct rewards can induce the same optimal policy, and potential-based shaping leaves optimal behaviour unchanged. Classical formulations resolve this ambiguity by adding principles or constraints. Maximum margin methods [22] separate expert from alternatives by a large margin, while Maximum Entropy IRL (MaxEnt) [30] posits the maximum-entropy trajectory distribution consistent with expert feature expectations,

$$P(\tau | R) \propto \exp\left(\sum_{t=0}^{T-1} R(s_t, a_t)\right), \quad (3.4)$$

closely linking IRL to occupancy matching. These estimators are interpretable and strong in tabular settings with known dynamics, but they typically assume fully observed, largely unconfounded data and can struggle when features or T must be learned.

3.3 Adversarial Inverse Reinforcement Learning

Adversarial formulations first arose in imitation learning via GAIL [11], which trains a policy to fool a discriminator into classifying learner trajectories as expert. AIRL [7]

casts IRL as a related game in which the discriminator’s logit yields a reward signal optimised by reinforcement learning. A common parameterisation writes

$$D_{\theta}(s, a, s') = \frac{\exp(f_{\theta}(s, a, s'))}{\exp(f_{\theta}(s, a, s')) + \pi_{\phi}(a | s)}, \quad (3.5a)$$

$$r_{\theta}(s, a) = \log D_{\theta}(s, a, s') - \log(1 - D_{\theta}(s, a, s')), \quad (3.5b)$$

and decomposes

$$f_{\theta, \psi}(s, a, s') = r_{\theta}(s, a) + \gamma V_{\psi}(s') - V_{\psi}(s).$$

Under *state-only, stationary* rewards, the potential terms V_{ψ} encourage rewards that are invariant to the dynamics, which can aid transfer. In practice AIRL often presumes full observability and homogeneous demonstrations; when hidden variables shape expert behaviour, the learned reward can inherit spurious correlations that degrade generalisation, motivating the causal perspective below.

3.4 Causal Inference and Structural Models

Structural Causal Models (SCMs) [18, 20] describe variables X_i via structural equations

$$X_i = f_i(\text{Pa}(X_i), U_i), \quad (3.6)$$

where $\text{Pa}(X_i)$ are parents in a directed acyclic graph and U_i are exogenous noises. A confounder is an unobserved variable that causally influences multiple variables, creating statistical dependencies that do not reflect the mechanism of interest.

In our setting we posit a latent Z that affects both which states are visited and which actions are chosen:

$$Z \rightarrow S_t, \quad Z \rightarrow A_t.$$

Experts may differ in style, risk tolerance, or context; demonstrations draw from $\pi_E(a | s, z)$, but z is unobserved to the learner. If a learned reward $r(s, a)$ quietly absorbs Z -specific patterns, it can perform well in-distribution yet fail under shifts in Z .

An invariance principle helps. Across environments or latent contexts indexed by Z , the mechanisms that encode true preferences should remain stable. Building on invariant prediction [19], our method (Chapter 4) biases learning toward rewards whose statistics are stable across Z while retaining AIRL’s adversarial structure. Concretely, we condition the discriminator on a latent z and regularise the $\text{per}(s, a)$ variability

of the learned reward across z ; the aim is to suppress confounder-specific shortcuts without discarding potential-based shaping.

All experiments are conducted in fully observed, tabular GridWorlds with known dynamics; Z is simulated and unobserved to the learner. This controlled setting isolates reward-learning effects and lets us probe robustness to confounding and generalisation to held-out regions without conflating results with model misspecification. Broader implications and limitations are discussed in Chapters 6–7.

3.5 Notation Summary

Symbol	Meaning
$\mathcal{S}, \mathcal{A}$	State and action spaces
$T(s' s, a)$	Transition kernel
$R(s, a)$	Reward function (to be learned)
γ	Discount factor
$\pi_\phi(a s)$	Stochastic policy with parameters ϕ
$V^\pi(s), Q^\pi(s, a)$	Value and action-value functions (Eq. 3.1, 3.2)
$\rho^\pi(s, a)$	Discounted occupancy measure (Eq. 3.3)
$D_\theta(s, a, s')$	AIRL discriminator; r_θ defined (Eq. 3.5)
$f_{\theta, \psi}, V_\psi$	AIRL decomposition terms
Z, z	Latent confounder (random variable and realisation)
$\text{Pa}(\cdot), U$	Parents and exogenous noise in SCMs (Eq. 3.6)

Table 3.1: Notation used throughout the dissertation.

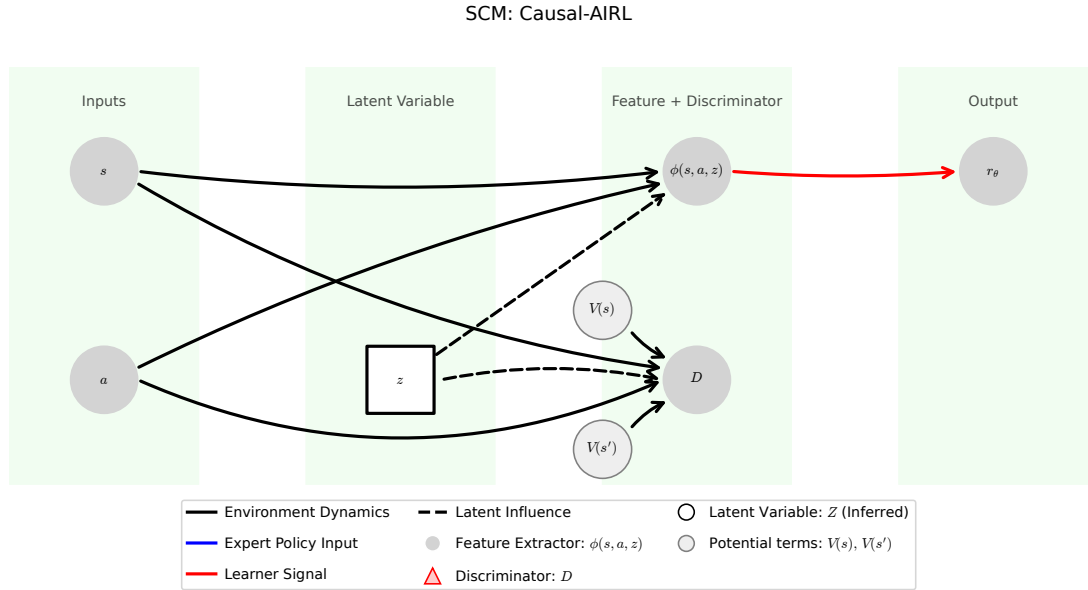
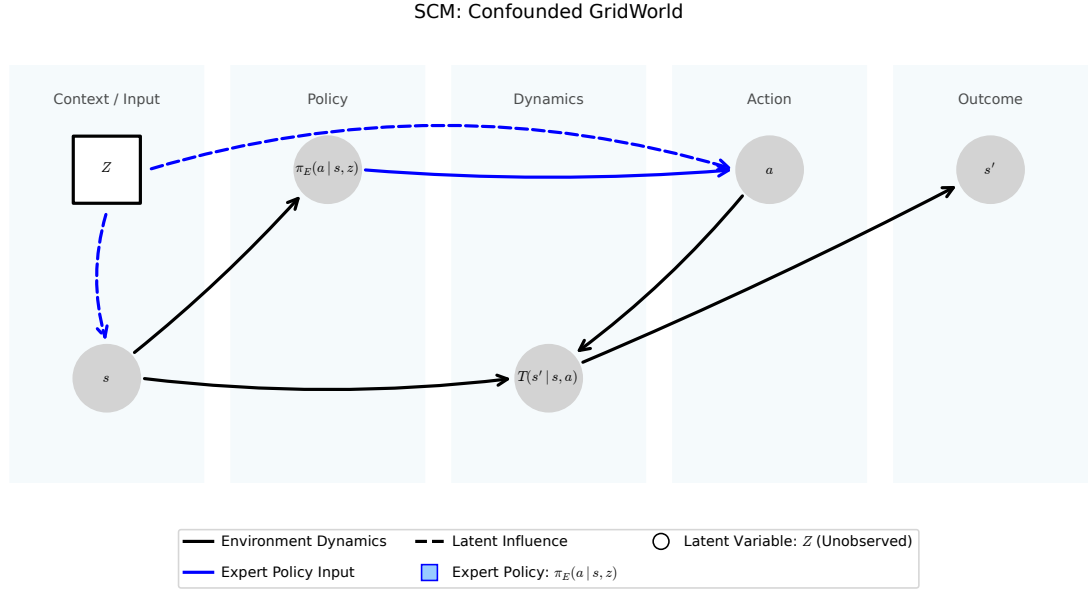


Figure 3.1: Structural models used in this work. Panel (a) depicts how an unobserved Z can simultaneously shape state visitation and action choice in expert data. Panel (b) shows the learner with latent conditioning and a bias toward Z -stable rewards.

Chapter 4

Methods

This chapter develops the learning objectives and modelling assumptions for the methods we evaluate: classical IRL baselines in the tabular regime, Adversarial IRL (AIRL), and the proposed Causal-AIRL. The presentation emphasises intuition first—what each objective is trying to achieve and why its assumptions matter—before stating the precise formulations used in our study. Throughout, we keep the setting and notation aligned with Chapter 3 and defer implementation minutiae and exhaustive hyperparameter listings to the Appendix to preserve the flow of the main text.

Notation. We write $\pi_\phi(a \mid s)$ for a stochastic policy with parameters ϕ , and use the occupancy notation of Eq. (3.3). All experiments in later chapters are conducted in fully observed, tabular GridWorld with known transitions T .

4.1 Classical IRL Baselines

Classical IRL formulates reward recovery as an inverse optimal-control problem with transparent constraints. In domains where dynamics are known and state spaces are modest, these methods remain interpretable anchors for evaluation: they make explicit how observed expert occupancies constrain admissible rewards, and they expose the central ambiguity of IRL—that many rewards can rationalise the same behaviour—so that any additional inductive bias is easy to scrutinise.

4.1.1 Ng and Russell (Projection IRL)

The projection formulation encodes the requirement that, under the learned reward, the expert’s feature visitation should dominate that of any competing policy. With

linear rewards $R(s) = \theta^\top \phi(s)$ (state features; the action-feature case is analogous), the feasibility constraints are

$$\theta^\top \mu_E \geq \theta^\top \mu^\pi \quad \forall \pi, \quad \mu^\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t) \right], \quad (4.1)$$

and one then searches within this feasible set for a reward that maximises the margin to the next-best policy. In small tabular MDPs this is attractive: the constraints are clear, optimisation is fast, and the resulting reward is easy to inspect. The same transparency highlights limitations: the solution is non-unique, it presumes access to T , and scaling is poor when the state space or feature representation grows. Occupancy measures (Eq. 3.3) provide an empirical route to estimating μ^π from trajectories.

4.1.2 Maximum Entropy IRL (MaxEnt)

Maximum Entropy IRL resolves underdetermination by favouring the most uncertain explanation consistent with the demonstrations. It posits that, among all distributions that match the expert’s feature expectations, the expert follows the maximum-entropy trajectory distribution,

$$P(\tau \mid R) \propto \exp \left(\sum_{t=0}^{T-1} R(s_t, a_t) \right). \quad (4.2)$$

With linear rewards $R(s, a) = \theta^\top \phi(s, a)$, matching features is equivalent to matching occupancies, providing a probabilistic estimator that tolerates stochasticity and mild suboptimality. In tabular settings with known T , MaxEnt yields explicit rewards and connects naturally to the occupancy view used throughout this thesis. When T is known, the partition function implied by Eq. (4.2) can be computed by soft value iteration, making tabular MaxEnt tractable in our setting.

For evaluation parity across baselines and deep methods, we compute the same metrics—primary reward fidelity via Spearman rank correlation, secondary value correlation and policy agreement—and we visualise reward maps identically. This keeps comparisons focused on objectives rather than on measurement artefacts.

4.2 Adversarial Inverse Reinforcement Learning

Building on GAIL [11], AIRL [7] casts IRL as an adversarial game between a policy (the generator) and a discriminator that distinguishes expert from learner rollouts. The discriminator’s logit furnishes a shaped reward that the policy then optimises, creating

a tight feedback loop between distinguishing signal and control. This formulation replaces hand-crafted features with learned representations while retaining an explicit reward usable beyond the training policy.

4.2.1 Objective and Reward

The discriminator assigns high scores to expert-like samples and low scores to learner samples. Writing its output and the induced reward as

$$D_{\theta}(s, a, s') = \frac{\exp(f_{\theta}(s, a, s'))}{\exp(f_{\theta}(s, a, s')) + \pi_{\phi}(a | s)}, \quad (4.3a)$$

$$r_{\theta}(s, a) = \log D_{\theta}(s, a, s') - \log(1 - D_{\theta}(s, a, s')), \quad (4.3b)$$

the discriminator is trained with binary cross-entropy on expert and policy rollouts, while the policy is updated by a simple REINFORCE rule with a learned baseline, advantage normalisation, entropy regularisation, and gradient clipping. This construction ties the reward directly to the signal that best separates expert from learner, a useful property when features are unknown and we want an explicit reward function.

4.2.2 Decomposition and Transfer

AIRL encourages dynamics-invariant rewards under *state-only*, *stationary* rewards through a potential-based decomposition:

$$f_{\theta, \psi}(s, a, s') = r_{\theta}(s, a) + \gamma V_{\psi}(s') - V_{\psi}(s). \quad (4.4)$$

The potential terms V_{ψ} shape learning without altering optimal policies, so when true rewards are state-only and stationary, the discriminator’s preference can align with a reward that transfers across dynamics. In our scope this decomposition also serves a second purpose: it guards against degenerate constant-reward solutions by separating policy-shaping potentials from the state–action reward term.

Shared stabilisers. We use a small entropy bonus to keep exploration alive and gradient clipping to prevent rare large updates from destabilising training. These guardrails are light-touch and are held fixed across AIRL and Causal-AIRL for fairness (cf. Table 4.1).

4.3 Causal-AIRL

Causal-AIRL augments AIRL with an explicit treatment of latent confounding. The motivation follows the structural picture in Fig. 3.1: expert trajectories may be influenced by an unobserved variable Z that affects both which states are visited and which actions are chosen. A discriminator that ignores Z can therefore ascribe Z -specific artefacts to the reward. Our design tackles this in two coordinated steps: latent conditioning, which exposes the discriminator to Z -dependent variation, and an invariance regulariser, which discourages reward fluctuations across Z at the same (s, a) .

4.3.1 Latent Conditioning

We equip the discriminator with a latent variable and condition it on an inferred z . Concretely, an inference network $q_\phi(z | s, a)$ approximates a posterior over latent contexts with prior $p_\omega(z)$, and the discriminator becomes

$$D_\theta(s, a, z) = \frac{\exp(f_\theta(s, a, z))}{\exp(f_\theta(s, a, z)) + \pi_\phi(a | s)}, \quad f_{\theta, \psi}(s, a, s', z) = r_\theta(s, a, z) + \gamma V_\psi(s') - V_\psi(s). \quad (4.5)$$

During training we draw $z \sim q_\phi(z | s, a)$ so that the discriminator learns to attribute Z -driven differences to the latent rather than the reward term; the prior $p_\omega(z)$ is taken as a standard normal. At evaluation we form Monte Carlo estimates with a small, fixed number of samples ($K=5$) or marginalise when closed forms exist, aligning the protocol with how invariance is measured in Chapter 5.

4.3.2 Invariance-Regularised Objective

Conditioning alone does not enforce stability. We therefore bias learning towards Z -stable rewards by augmenting the discriminator-side objective with two terms:

$$\mathcal{L}_{\text{KL}} = \mathbb{E}_{(s, a)} [\text{KL}(q_\phi(z | s, a) \| p_\omega(z))], \quad (4.6)$$

$$\mathcal{L}_{\text{inv}} = \mathbb{E}_{(s, a)} [\text{Var}_{z \sim q_\phi}(r_\theta(s, a, z))], \quad (4.7)$$

and the total discriminator objective

$$\mathcal{L}_D^{\text{CAIRL}} = \mathcal{L}_D^{\text{AIRL}} + \lambda_{\text{kl}} \mathcal{L}_{\text{KL}} + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}}. \quad (4.8)$$

In practice $\text{Var}_{z \sim q_\phi}(r_\theta(s, a, z))$ is estimated with the unbiased sample variance using $K=5$ samples; the same K is used at evaluation for invariance metrics to keep train/test

consistent. The KL term keeps the latent posterior well-behaved (optionally warmed up over early epochs), while the invariance penalty directly discourages per- (s, a) reward fluctuations across Z . Crucially, this operates alongside AIRL’s potential-based decomposition: the policy still learns from $r_\theta(s, a, z)$, but the learned reward is nudged toward a representation that does not rely on confounder-specific shortcuts. A mild clamp on $f_{\theta, \psi}$ before the logistic transform in D is used for numerical hygiene and does not change the objective’s semantics.

Why this targets reward robustness. If the expert’s underlying preferences are stable across contexts, then two trajectories that differ only by Z should be explained by the same reward signal at the same (s, a) . Minimising Eq. (4.7) implements this idea directly. When Z is irrelevant (our clean regimes), the penalty is effectively idle and the method behaves like AIRL with a light additional regularisation; when Z matters (our confounded and held-out suites), the penalty suppresses the very fluctuations that would otherwise be mistaken for preference.

4.4 Implementation Notes and Guardrails

To ensure comparisons are informative rather than artefactual, we adopt a simple protocol. First, we calibrate AIRL in clean GridWorld by selecting modest values for the shared stabilisers (entropy bonus and gradient clipping) and then lock these values. Second, we introduce Causal-AIRL under the same training recipe and tune only its causal-specific weights (the coefficients on the KL and invariance terms and the latent dimensionality) within the confounded setting; once chosen, these are likewise fixed across all experiment families. This avoids per-protocol retuning and keeps differences attributable to the objectives themselves. The network architectures are minimal (two-layer MLPs with Adam) to match the tabular nature of the tasks; classical IRL uses planners with known T . The policy is updated by REINFORCE with a learned baseline, advantage normalisation, entropy regularisation, and gradient clipping. Exact settings and best hyperparameters are listed in the Appendix for reproducibility; here we note only that the number of latent samples used to estimate the variance term is kept small and constant ($K=5$) to bound the computational overhead and to align training and evaluation.

Table 4.1 summarises the small set of training knobs and their roles. The shared entries appear in both AIRL and Causal-AIRL; the causal-specific entries control the latent posterior’s hygiene and the strength of the invariance bias. Using K samples for

$\mathcal{L}_{\text{inv}}$ adds an $O(K)$ factor to the discriminator-side compute; in our experiments this remains modest and is reflected in the scaling curves of Chapter 6.

Table 4.1: Training knobs (names match configs).

Knob	Role	Where used
entropy_coef	Policy entropy bonus	AIRL, Causal-AIRL
grad_clip_norm	Stabilise policy updates	AIRL, Causal-AIRL
reward_logit_clip	Clamp $f_{\theta, \psi}$ before logistic transform	AIRL, Causal-AIRL
kl_coeff, kl_warmup_epochs	Posterior hygiene (ramp)	Causal-AIRL
inv_coeff	Reward invariance weight	Causal-AIRL
latent_dim	Latent dimensionality d_Z	Causal-AIRL
num_z_samples	MC samples K for $\mathcal{L}_{\text{inv}}$	Causal-AIRL

Assumptions for this study are simple and shared across methods. The models are instantiated and compared in fully observed GridWorlds with known dynamics; Z is simulated and unobserved to the learner; there is no interventional access to alter Z . Evaluation uses the same metrics across methods (primary Spearman, secondary value/policy; robustness via reward-variance across Z and worst- Z overlays). Taken together, this chapter instantiates Causal-AIRL as a minimal extension of AIRL—adding latent conditioning and an invariance penalty while holding shared stabilisers fixed and using the same REINFORCE update—so that differences in the results can be attributed to how the objectives treat latent variation rather than to architectural or optimisation confounds.

Chapter 5

Experiments

This chapter describes the empirical design used to compare Ng–Russell, MaxEnt, AIRL, and Causal-AIRL. Our aim is to test reward recovery and transfer in a controlled, fully observed tabular setting while varying the factors that most plausibly affect robustness: data availability, transition noise, latent confounding, held-out regions, and problem size. We use small grids deliberately to prioritise statistical reliability over a few long runs and to benchmark fairly against classical tabular methods; conclusions are drawn from aggregates over repeated trials rather than from isolated anecdotes. All figures referenced here appear in Chapter 6; this chapter focuses on what is varied, why it is varied, and how the measurements are computed.

5.1 Environment and controllable factors

Experiments take place in a finite GridWorld MDP with actions $\{\text{up}, \text{down}, \text{left}, \text{right}\}$. Dynamics are known and tabular. Controlled stochasticity is introduced via slip: with probability $p \in \{0, 0.1, 0.2\}$, the intended action is replaced uniformly at random by one of the other three actions. This simple setting exposes core trade-offs (credit assignment, noise sensitivity, and transfer across contexts) without confounding them with perception or model-misspecification. Environment layouts and transition details are summarised in A.

We study two reward designs that bracket common regimes. Sparse rewards (non-zero only at terminal states) induce hard credit assignment, whereas shaped rewards provide potential-based guidance toward the goal. Keeping both allows us to check whether any apparent gains rely on shaping.

Expert trajectories are produced by entropy-regularised planning on the true

MDP, yielding soft-optimal policies. We vary the number of demonstrations $N \in \{5, 10, 20, 50\}$ to probe data sparsity. In confounded suites (Section 5.4), the expert policy depends on a latent variable $Z \in \{0, 1\}$ (for example, different styles or constraints), producing two subpopulations whose traces are indistinguishable to a learner that ignores Z .

5.2 Evaluation questions and families

Our experiments are organised into four families that answer complementary questions. First, clean sensitivity sweeps vary demonstrations, slip, reward type, and discount factor $\gamma \in \{0.9, 0.95, 0.99\}$ under the usual IRL assumptions (no confounder, known dynamics) to establish parity on benign problems and to expose sensitivity to data and stochasticity. Second, confounded-expert protocols train on one Z (or mixtures) and evaluate both in-distribution and across Z , isolating the central claim about latent confounding. Third, held-out generalisation applies a spatial mask during training and then evaluates both on the training support and in the masked region, checking whether methods extrapolate structure or overfit context-specific artefacts. Finally, scaling traces accuracy versus wall-clock time as grid size increases, emphasising relative accuracy–time frontiers rather than absolute times.

Table 5.1: Experiment families. What varies, what is held fixed, and the primary question each family answers.

Family	Varies	Fixed	Question
Clean sweeps	N , p , reward type, γ	No confounder; known T	Benign-regime fidelity and stability under data/noise.
Confounded experts	Train/test Z	$p=0$ unless stated; identical eval seeds	Does Z -aware invariance improve cross- Z transfer?
Held-out generalisation	Train mask, train/test Z	Same mask across methods	Do rewards extrapolate structure into unvisited regions?
Scaling	Grid size N	Method settings fixed from sweeps	Accuracy–time frontier and overheads.

5.3 Metrics and aggregation

Primary fidelity is measured by Spearman rank correlation between the learned reward $r(s, a)$ and the ground-truth reward $R_{\text{true}}(s, a)$. Because IRL is identifiable only up to monotone transformations, a rank statistic avoids misleading scale effects and keeps comparisons fair across parameterisations.

As secondary checks we report (i) value correlation, taken as the Pearson correlation between induced value functions, and (ii) policy agreement, the fraction of action matches between learned and expert policies on identically seeded evaluation rollouts. Identical seeds reduce evaluation noise but can compress apparent variability; conclusions are therefore based on shifts in means rather than isolated outliers.

To summarise robustness to latent context, we compute the across- Z reward variance $\text{Var}_z(r(s, a, z))$, estimated with the unbiased sample variance using $K=5$ latent samples per (s, a) , and then average uniformly over (s, a) . Lower values indicate a reward less sensitive to latent context at the same state–action.

For compute we record wall-clock time per run on fixed hardware and plot accuracy versus time; absolute times are hardware dependent, so we emphasise relative frontiers. Hardware and timing protocol appear in D.

Unless noted otherwise, each configuration is repeated with five independent seeds. We report the mean and a 95% confidence interval obtained by bootstrap (1,000 resamples with replacement). When a configuration has $n=1$, the interval reduces to a point. For each metric we use the final finite value at convergence, and correlations are clipped to $[-1, 1]$ for numerical hygiene.

5.4 Setups in detail

Clean sensitivity sweeps

We vary demonstrations N , slip p , reward type (sparse versus shaped), and γ , holding everything else fixed and excluding confounders. This answers two questions: whether Causal-AIRL pays a cost in benign regimes, and how sensitive each method is to data and noise. Stability knobs (entropy bonus and gradient clipping) are chosen once on clean grids and then reused unchanged across methods and families; Causal-AIRL uses its default latent settings; we fix the latent estimator at $K = 5$ for all reported plots (ablations with $K \in \{1, 5\}$ showed negligible effect on clean-regime metrics and are

omitted).

Confounded experts and cross- Z transfer

We simulate a latent confounder $Z \in \{0, 1\}$ that influences both state visitation and expert action choice. Training is performed on trajectories generated with one Z (or mixtures), and evaluation is performed both in-distribution and across Z (for example, train on $Z=0$, test on $Z=1$ and vice versa). We vary demonstration counts $N \in \{10, 20, 40\}$ in this suite. Policy agreement under such cross- Z evaluation summarises transfer without assuming calibrated reward scales. Invariance is measured on matched scenarios where exactly one run per method per Z is available, making the variance comparison meaningful.

Generalisation to held-out regions

To test extrapolation, we define a contiguous spatial mask and exclude those states during training. Masks are quadrant-shaped $\{top_left, top_right, bottom_left, bottom_right\}$ and are shared across methods within each run. Evaluation then proceeds both on the training support and within the masked region. Overlays use a shared normalisation and colormap across methods so that visual comparisons reflect substantive differences. When combined with cross- Z evaluation, this protocol emphasises whether rewards capture structure that survives both coverage gaps and latent shifts.

Scaling

We measure accuracy versus wall-clock time as grid size increases ($N \times N$ with $N \in \{5, 7, 9, 11, 13, 15\}$; extended where reported). Classical IRL is particularly strong at these scales (exact planners and convex programs remain tractable), so this setting provides a fair accuracy–time baseline. For Causal-AIRL, latent sampling uses a small, fixed number of draws ($K=5$) to bound overhead and to match the invariance estimator in Section 5.3. Unless stated otherwise, scaling uses sparse rewards, $p = 0$, and trains for `irl.max_iters` = 200 per run.

5.5 Method-specific instantiation

Classical IRL (Ng–Russell; MaxEnt) assumes known dynamics and operates in the tabular regime. We present these methods as interpretable anchors: their objectives directly encode optimality or feature-expectation constraints, and the learned rewards can be inspected state by state. They are evaluated with the same metrics and visualisations as the deep methods.

AIRL is trained with a small entropy bonus (to maintain exploration) and gradient clipping (to prevent rare large updates), then held fixed across families. The discriminator uses the standard potential-based decomposition so that policy-shaping potentials are separated from the reward term.

Causal-AIRL augments AIRL with a latent variable z and an invariance penalty on the per- (s, a) variability of $r(s, a, z)$ across z . The latent posterior is regularised with a mild KL term, optionally warmed up early in training for numerical hygiene. Importantly, the policy update is unchanged—it still optimises the discriminator-derived reward—so differences arise from how the reward is identified in the presence of latent variation. To keep comparisons clean, shared stabilisers match AIRL’s, and causal-specific coefficients are selected once in the confounded suite and reused elsewhere.

5.6 Sanity checks and failure guards

Before large sweeps, we verify on 3×3 and 4×4 deterministic grids with hand-crafted rewards that (i) learned rewards peak near the goal when they should, (ii) MaxEnt matches feature expectations, and (iii) induced values increase with proximity to the goal. During training, we monitor discriminator and policy losses to rule out degenerate constant-logit solutions that could artificially reduce variance. All methods share the same evaluation seeds in a given configuration so that policy agreement reflects learned differences rather than roll-out noise.

5.7 Assumptions, scope, and reproducibility

The study assumes fully observed states, known tabular dynamics, and a simulated but unobserved confounder Z . This isolates reward-learning effects and keeps the comparison with classical baselines honest. Where matched-slice summaries have $n=1$, we treat them as qualitative checks aligned with cross- Z bars rather than universal claims.

Full configuration listings, hardware notes, and scripts to regenerate figures appear in the A; the main text remains method-focused and result-driven. In this sense, the contribution of this chapter is an evaluation design that isolates confounding, generalisation, clean-regime parity, and scaling, with metrics aligned to reward-level fidelity and invariance and with protocol choices that keep comparisons statistically reliable and fair.

Chapter 6

Results

We evaluate Ng–Russell, MaxEnt, AIRL, and **Causal-AIRL** in tabular GridWorld across four families—*clean sweeps*, *confounded*, *generalisation to held-out states*, and *scaling*—using **reward Spearman** as the primary fidelity metric (higher is better) with *value correlation* and *policy agreement* as secondary checks, and an invariance summary given by per- (s, a) reward variance across Z (lower is better). Unless noted we report means over five seeds with 95% CIs; when $n=1$ the interval reduces to a point. Because reward is only identifiable up to monotone transforms, rank correlation is preferred to scale-sensitive metrics. Classical IRL (Projection IRL; MaxEnt) provides tabular anchors; AIRL is the deep adversarial comparator; Causal-AIRL augments AIRL with latent Z -conditioning and a penalty on the per- (s, a) reward variance across Z . We probe robustness by varying reward density, transition noise (slip), and confounding. Figures follow a concise, figure-first style; text paraphrases trends without fragile point estimates, and full interpretation is reserved for Chapter 7. Unless otherwise stated, all results in this chapter are obtained in fully observed, tabular GridWorlds with known transition dynamics T and a simulated latent confounder Z .

6.1 Classical IRL (Ng–Russell, MaxEnt)

In clean GridWorlds without confounders, classical tabular methods behave as expected: Spearman improves monotonically with more demonstrations and tighter shaping, providing calibration anchors rather than primary claims. AIRL on clean maps mirrors these trends and supplies a deep reference against which Causal-AIRL deltas under shift are read.

6.2 AIRL (reference comparator)

The confounded and generalisation protocols mirror those used for Causal-AIRL to enable direct comparison. AIRL’s cross-Z transfer is sensitive to confounding, motivating an explicit Z-aware invariance bias.

6.3 Causal-AIRL

We begin with cross-Z transfer under confounding. Agreement is a robust summary when reward scales are incomparable, and we complement it with an invariance statistic that aggregates per- (s, a) variance over $K=5$ latent draws z and then averages uniformly over (s, a) to avoid visitation confounds.

We first test whether learned rewards *transfer across latent contexts*. We report cross-Z *policy agreement* (higher is better) and a reward-level *across-Z variance* summary (lower is better). Evaluation rollouts are identically seeded to reduce noise, which tightens confidence intervals.

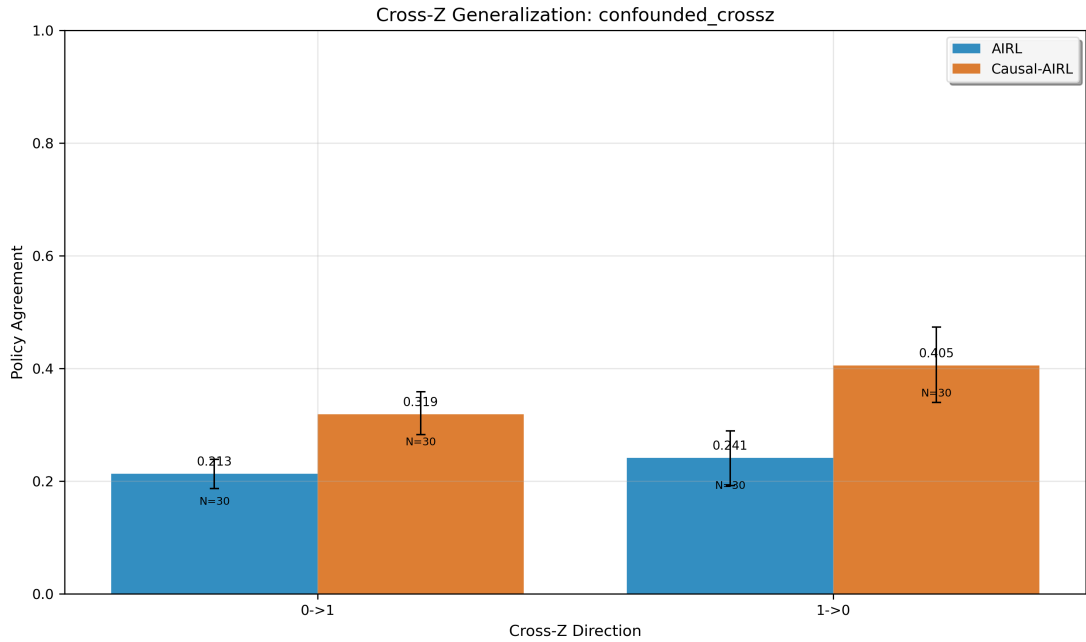


Figure 6.1: Confounded GridWorld: cross-Z policy agreement ($0 \rightarrow 1$, $1 \rightarrow 0$). Bars show mean with 95% bootstrap confidence intervals (higher is better).

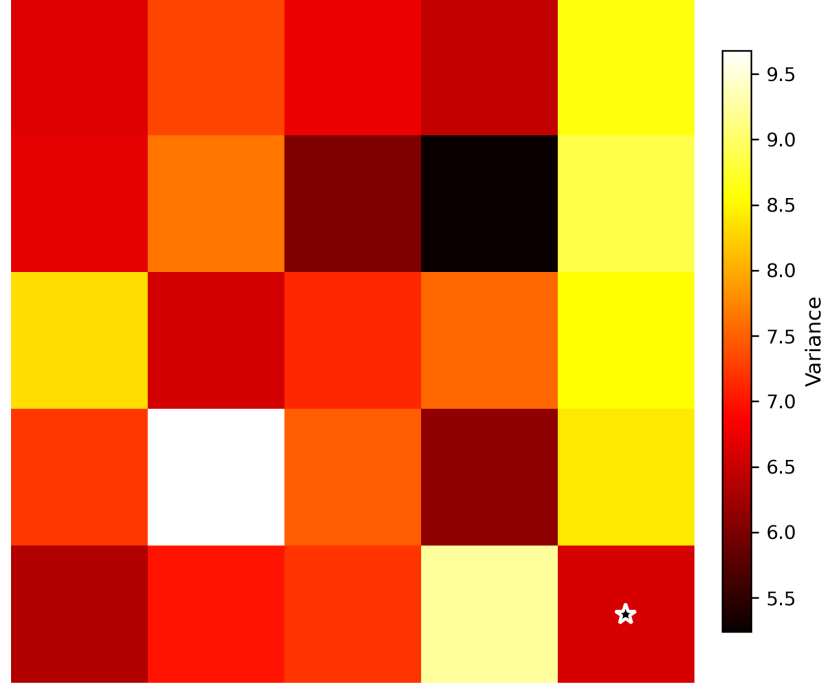
AIRL— $0 \rightarrow 1$: 0.213 [0.187, 0.239], $1 \rightarrow 0$: 0.241 [0.192, 0.289];

Causal-AIRL— $0 \rightarrow 1$: 0.319 [0.283, 0.359], $1 \rightarrow 0$: 0.405 [0.340, 0.473].

Conditioning on Z with an invariance penalty improves cross-Z transfer in both directions.

Relative to AIRL, Causal-AIRL yields consistently higher cross-Z policy agreement in both transfer directions (Fig. 6.1). The asymmetry between $0 \rightarrow 1$ and $1 \rightarrow 0$ persists but is modest and aligns with map topology (corridor and obstacle placement) rather than optimiser noise; evaluation rollouts are identically seeded and measured at convergence, so differences reflect learned policies rather than sampling artefacts.

Reward Invariance Across Z — mode=across_runs (groupby=env.confounder_value)
Values: 0, 1 — Worst-Z PolicyAgree=0.295 (Mean Variance: 7.3510)



	Mean variance	n maps
Across Z	7.35	90
AIRL	6.02	3
Causal-AIRL	5.31	3

Figure 6.2: Confounded GridWorld: reward variance across Z (lower is better). We report the per- (s,a) variance of the learned reward across Z, averaged uniformly over the grid. Mean variance across all runs is 7.35 (90 maps). On matched scenarios (one run per method per Z), AIRL averages 6.02 vs. Causal-AIRL 5.31. The accompanying worst-Z policy agreement is 0.295. Lower variance for Causal-AIRL is consistent with suppressing confounder-specific fluctuations at salient (s,a) rather than collapsing variance globally.

The variance summary in Fig. 6.2 targets invariance of the *reward* itself, not just the induced policy. The observed mean variance (7.35 across all runs) and worst-Z policy

agreement (0.295) indicate that across- Z fluctuations remain but are structured. Per-method comparison highlights that Causal-AIRL attains lower reward variance (5.31 vs. AIRL 6.02), reinforcing the interpretation that robustness arises from dampening spurious Z -specific variation rather than collapsing reward diversity. Degenerate constant-logit behaviour, which would trivially minimise variance, is contradicted by the training diagnostics in Fig. 6.9.

We next probe *held-out generalisation*: training excludes a spatial region, then we evaluate both on the training support and in the *held-out region*, across Z . This asks whether rewards extrapolate structure rather than encode context-specific artefacts.

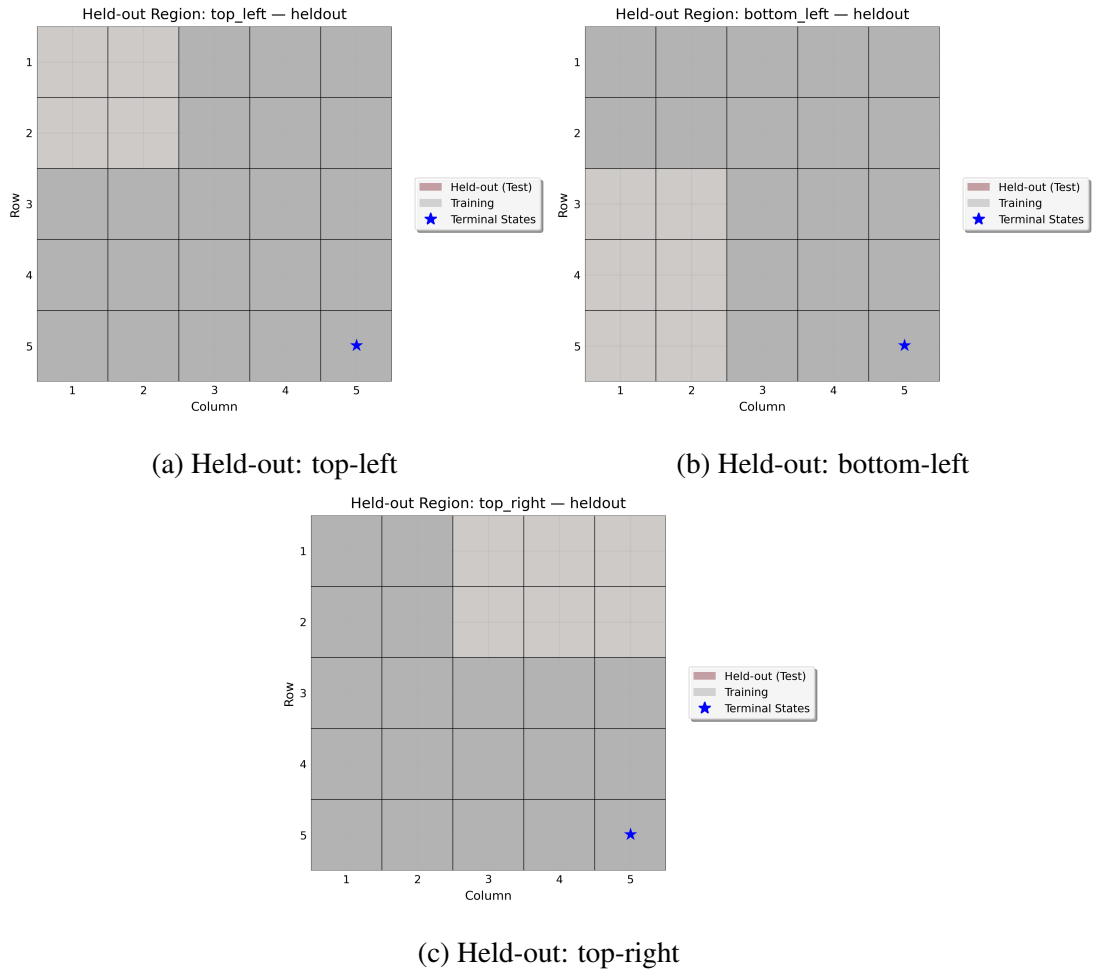


Figure 6.3: Generalisation masks: held-out region layouts (qualitative). Shaded regions indicate states excluded during training and reserved for evaluation. Panels serve only to clarify the evaluation setup; no scalar comparisons should be drawn from textures.

Figure 6.3 illustrates example held-out regions used in the generalisation protocol. These spatial masks define which portions of the grid were excluded during training and subsequently reserved for evaluation. The layouts are included for context only,

as they clarify the evaluation setup without themselves conveying quantitative differences.

The corresponding quantitative results are reported in Fig. 6.4. There, cross-Z agreement is measured under both confounded and held-out protocols. The observed gains for Causal-AIRL relative to AIRL persist across all conditions, indicating that the benefits of conditioning with an invariance penalty extend beyond the confounded training regime and into explicit held-out generalisation.

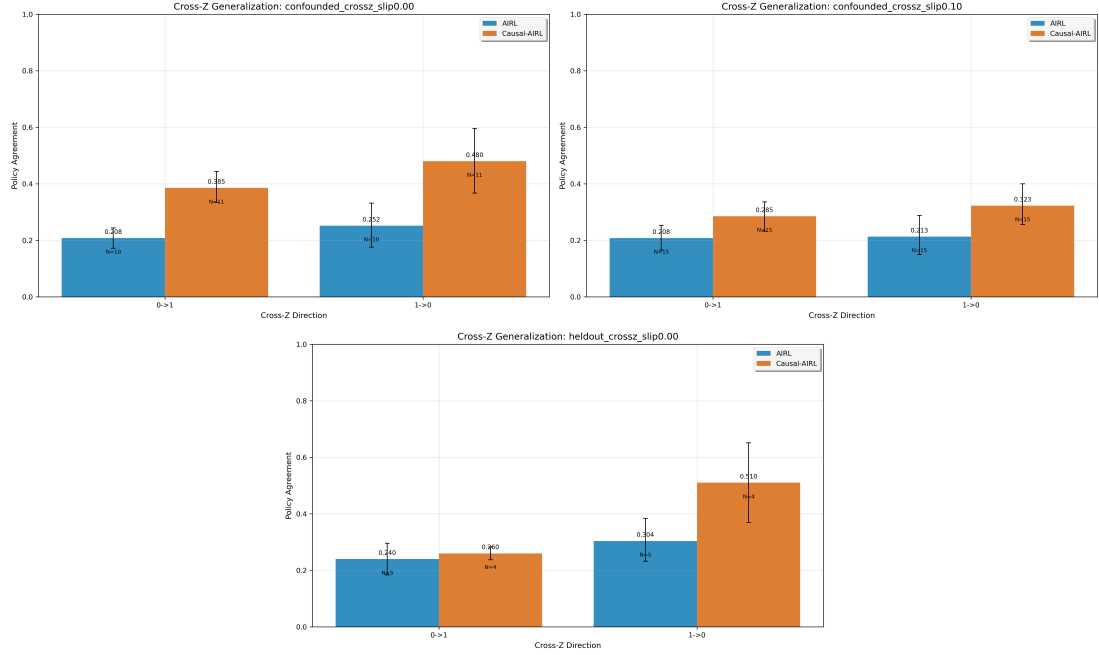


Figure 6.4: Generalisation protocol: cross-Z policy agreement ($0 \rightarrow 1$, $1 \rightarrow 0$) under confounded and held-out variants. Means with 95% bootstrap confidence intervals (higher is better).

(a) Confounded, slip $p=0.0$:

AIRL $0 \rightarrow 1$: 0.208 [0.172, 0.244], $1 \rightarrow 0$: 0.252 [0.176, 0.332];

Causal-AIRL $0 \rightarrow 1$: 0.385 [0.335, 0.444], $1 \rightarrow 0$: 0.480 [0.367, 0.596].

(b) Confounded, slip 0.1:

AIRL $0 \rightarrow 1$: 0.208 [0.165, 0.253], $1 \rightarrow 0$: 0.213 [0.149, 0.288];

Causal-AIRL $0 \rightarrow 1$: 0.285 [0.232, 0.336], $1 \rightarrow 0$: 0.323 [0.256, 0.400].

(c) Held-out, slip 0.0:

AIRL $0 \rightarrow 1$: 0.240 [0.184, 0.296], $1 \rightarrow 0$: 0.304 [0.232, 0.384];

Causal-AIRL $0 \rightarrow 1$: 0.260 [0.237, 0.283], $1 \rightarrow 0$: 0.510 [0.369, 0.651].

Gains for Causal-AIRL persist under added slip and explicit held-out evaluation.

Relative to AIRL, Causal-AIRL consistently improves cross-Z agreement in the generalisation protocol (Fig. 6.4). With zero slip, the advantage is large in both direc-

tions; with slip $p = 0.1$, absolute performance drops for all methods but Causal-AIRL retains a margin. In the held-out setting, Causal-AIRL is comparable to AIRL on $0 \rightarrow 1$ but markedly better on $1 \rightarrow 0$ (0.510 vs. 0.304). These results indicate that the robustness benefits of conditioning with an invariance penalty extend beyond the confounded regime, persisting under explicit held-out evaluation.

6.4 Training diagnostics (context)

To ensure robustness does not come at the expense of benign-regime performance, we vary demonstrations N , slip p , reward type, and γ (others at defaults). Figs. 6.5–6.7 summarise reward fidelity and how it propagates to value and policy.

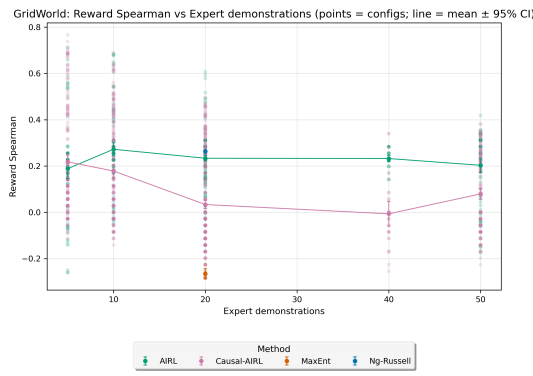
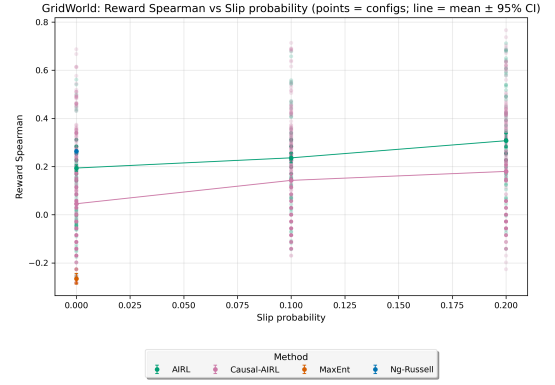
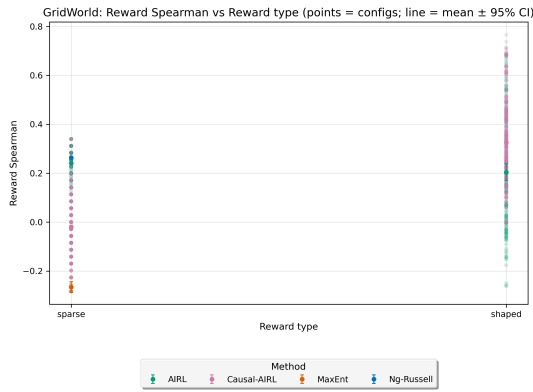
To check that robustness does not cost benign-regime fidelity, we sweep demonstrations N , slip p , reward type (sparse vs. shaped), and γ , holding other settings fixed. The primary metric is reward Spearman; value correlation and policy agreement are secondary.

Panel (a): AIRL benefits quickly from few demos then plateaus; Causal-AIRL tracks the trend at slightly lower means—extra regularisation offers no clean-regime gain. (b): Increasing slip mildly helps both (broader state coverage), keeping AIRL’s margin. (c): With *shaped* rewards, Causal-AIRL is best; with *sparse*, AIRL/Ng–Russell calibrate better. (d): Long-horizon emphasis is neutral for AIRL but interacts with Causal-AIRL’s invariance, reducing Spearman at $\gamma=0.99$.

Overall, clean sweeps confirm that the gains of Causal-AIRL are *not* from inflating clean-regime correlation; rather, its advantages emerge under latent shift and held-out evaluation (Figs. 6.1–6.4), while maintaining reasonable parity in benign conditions—especially with shaped rewards.

Value trends mirror Spearman: AIRL improves with shaping and slip; Causal-AIRL hovers near zero/negative. This does not contradict its policy robustness—value correlation is fragile to potential-shaping, scale, and γ . In benign settings, AIRL/Ng–Russell set the value baseline; Causal-AIRL trades some value correlation for the cross- Z gains in Figs. 6.1–6.4.

Across sweeps, AIRL generally leads on policy agreement, most clearly as slip increases and with shaped rewards; Causal-AIRL is closest under low slip and sparse rewards, and briefly exceeds AIRL at $N=40$. Both methods show little benefit from larger γ , with AIRL declining toward $\gamma=0.99$. Overall, the invariance penalty does not degrade policy quality in benign settings; it mainly reallocates capacity toward the

(a) Spearman vs demos N (b) Spearman vs slip p 

(c) Spearman vs reward type

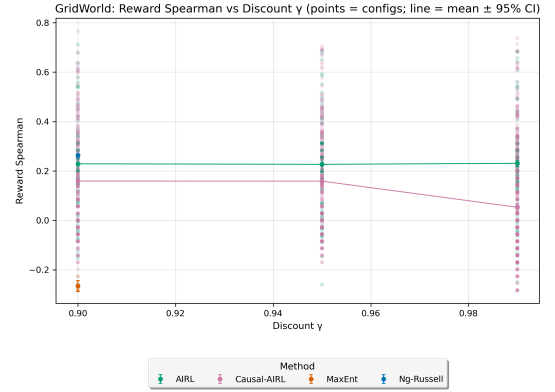
(d) Spearman vs γ

Figure 6.5: Clean sweeps: reward Spearman across demonstrations N , slip p , reward type, and γ (points = configurations; line = mean $\pm 95\%$ confidence interval). With more demonstrations, AIRL rises and plateaus while Causal-AIRL tracks slightly lower. Increasing slip improves both methods; shaped rewards help all methods and favour Causal-AIRL on Spearman, whereas sparse rewards favour AIRL/Ng–Russell. Higher γ is neutral for AIRL and mildly reduces Causal-AIRL, consistent with an invariance bias tuned for confounded regimes rather than maximising clean-regime rank correlation.

cross- Z robustness benefits established earlier. In sum: AIRL (and Ng–Russell) set the clean-regime baseline; Causal-AIRL is competitive on policy agreement (especially at low slip/sparse) and wins under shaped rewards on Spearman, but is modestly lower elsewhere and on value correlation—an expected trade-off for invariance. Crucially, near-parity in clean regimes means the robust cross- Z gains come without sacrificing benign performance.

Finally, we examine *accuracy–time trade-offs* and *training stability*. We show accuracy vs. wall-clock (log-scale) across grid sizes and summarise discriminator/policy diagnostics. Latent sampling uses $K=5$ for both training and the invariance estimator

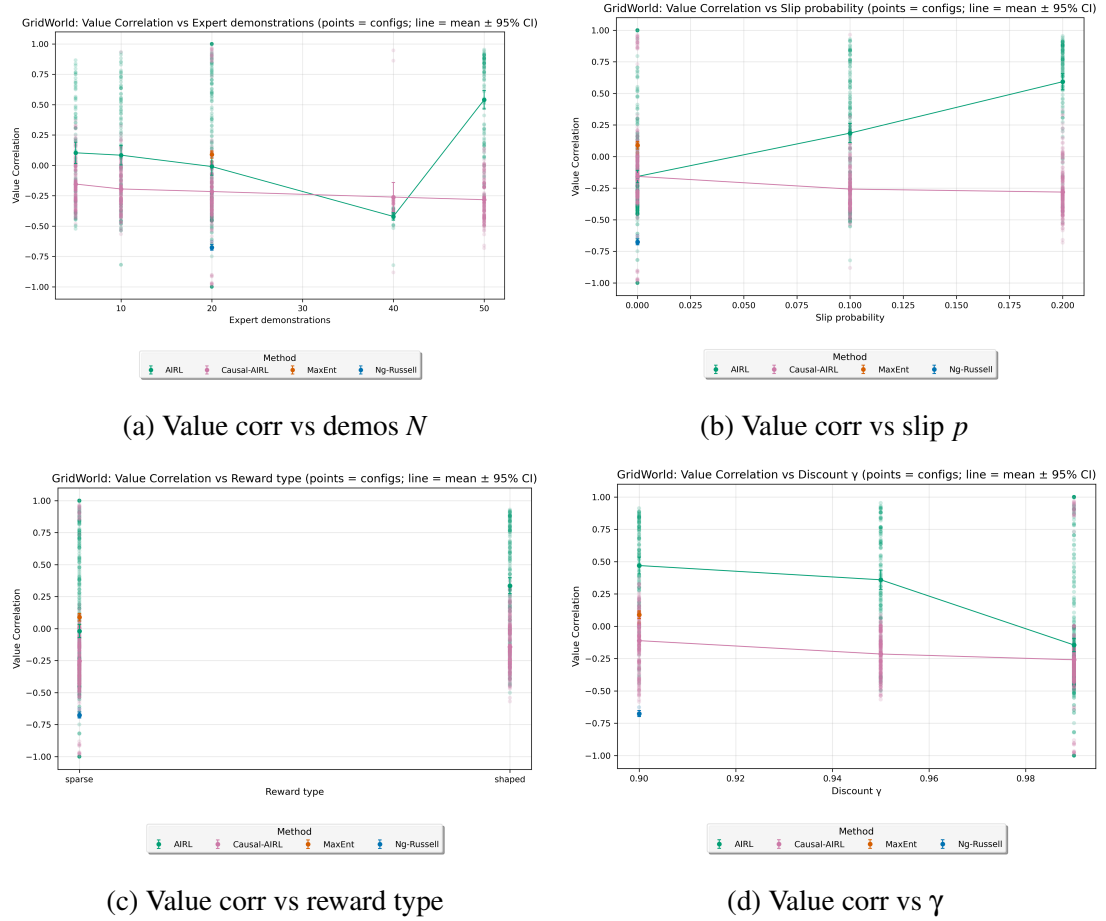


Figure 6.6: Clean sweeps: value correlation across N , p , reward type, and γ (points = configurations; line = mean $\pm$ 95% confidence interval). AIRL benefits from shaping and slip and dominates this secondary metric; Causal-AIRL remains near zero/negative in clean regimes. Because value correlation is sensitive to shaping, scale, and horizon, primary conclusions rely on reward Spearman and policy agreement.

(hardware and timing details in D).

Across sizes, accuracy increases with wall-time and exhibits a clear knee, after which gains taper. Causal-AIRL’s additional cost shifts curves slightly right but provides higher value correlation and non-decreasing Spearman as grids grow, while maintaining policy parity with AIRL. In follow-up runs (not shown), reducing $K \rightarrow 1$ recovers AIRL-like wall-time with a mild drop in invariance—matching expectations.

The losses follow the expected phases—burn-in, discriminator catch-up, and policy consolidation—without signs of discriminator overpowering. Environment-step counts fall gradually (improving efficiency), and Causal-AIRL’s posterior KL ramps then plateaus at a bounded level, evidencing non-collapsed latents. Policy loss stabilises earlier under Causal-AIRL, compatible with a less noisy reward signal. To-

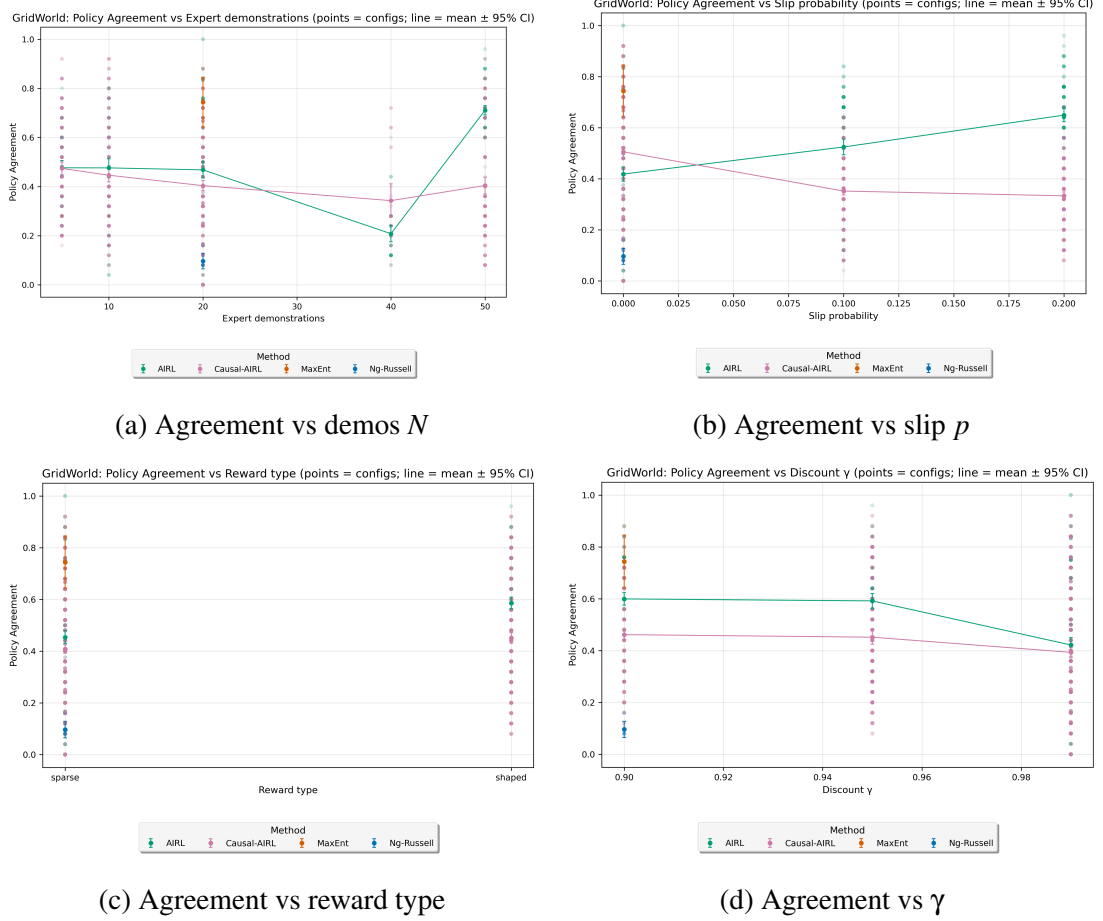
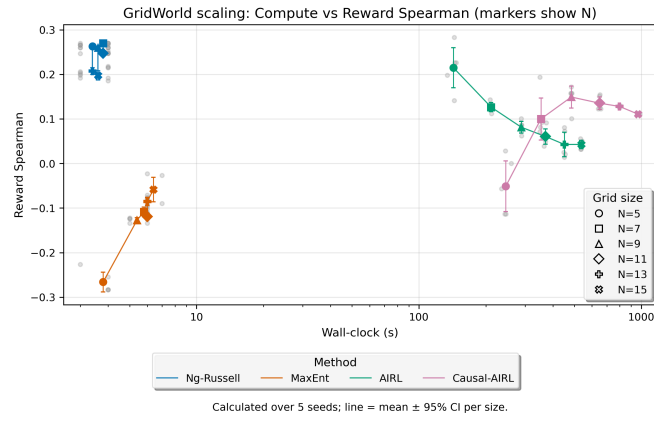


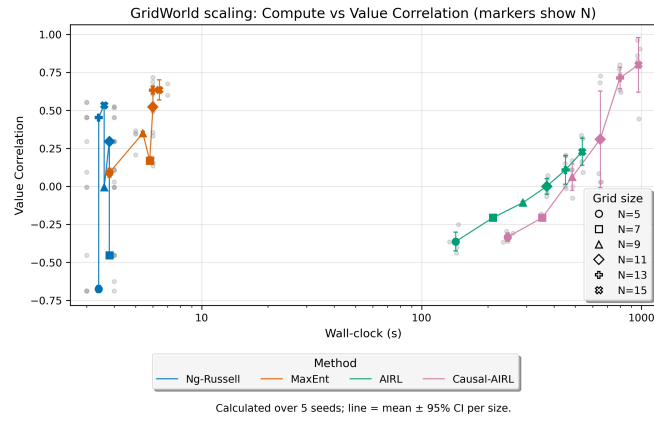
Figure 6.7: Clean sweeps: policy agreement across N , p , reward type, and γ (points = configurations; line = mean $\pm$ 95% confidence interval; higher is better). Agreement is broadly similar across AIRL and Causal-AIRL; AIRL tends to rise with slip and under shaped rewards, while Causal-AIRL is competitive at low slip and with sparse rewards. Invariance regularisation does not degrade policy quality in benign settings.

gether with our cross- Z and invariance results, these diagnostics support the claim that improvements come from suppressing Z -specific fluctuations rather than arbitrary reward rescaling.

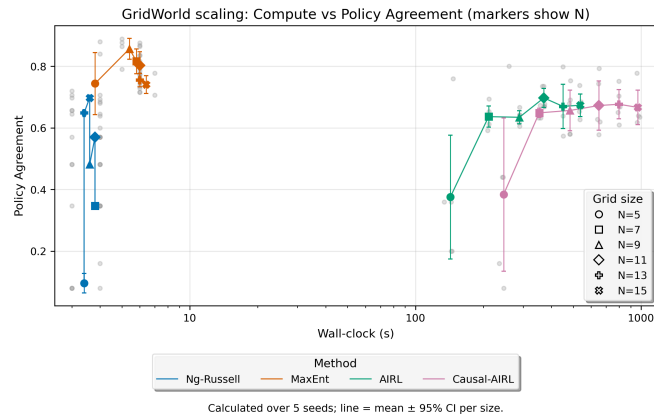
This chapter shows that Causal-AIRL improves cross- Z transfer and matches clean-regime fidelity with comparable compute, supporting invariance-regularised reward learning as a practical default when robustness to latent shift matters.



(a) Spearman vs wall-time

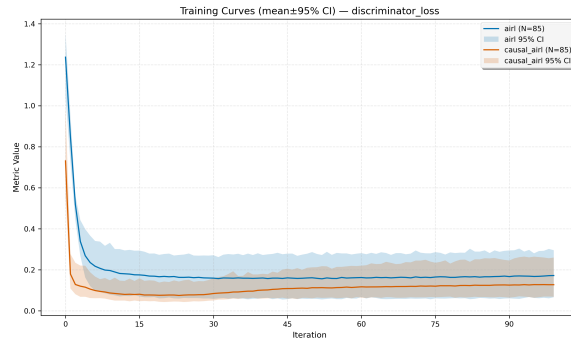


(b) Value corr vs wall-time



(c) Policy agreement vs wall-time

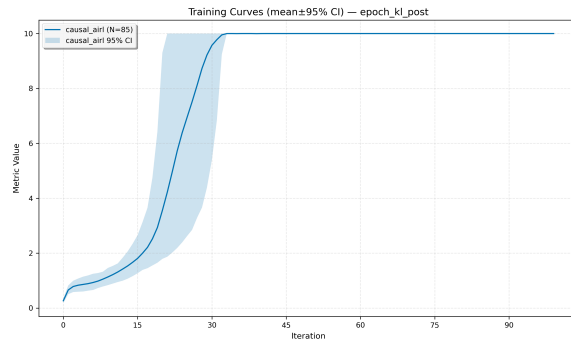
Figure 6.8: Scaling: accuracy vs. wall-clock time (log-scale x); grid sizes $N \in \{5, 7, 9, 11, 13, 15\}$, marker shape encodes N , lines show mean $\pm$ 95% confidence interval. Reward Spearman: AIRL trends down with compute while Causal-AIRL rises and overtakes at larger grids; value correlation: both improve, with Causal-AIRL climbing steeply; policy agreement: both rise then plateau. Causal-AIRL adds modest overhead (latent sampling $K=5$, extra discriminator terms) yet remains close to AIRL on the accuracy–time frontier; absolute times are hardware-dependent, so relative trends matter.



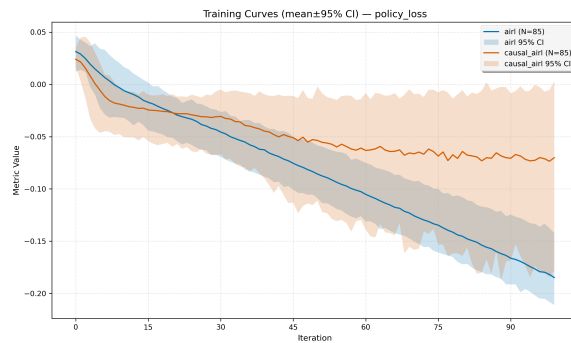
(a) AIRL and Causal-AIRL: discriminator loss



(b) AIRL and Causal-AIRL: environment steps



(c) Causal-AIRL: posterior KL per epoch



(d) AIRL and Causal-AIRL: policy loss

Figure 6.9: Training diagnostics (mean $\pm$ 95% confidence interval). Discriminator loss drops and stabilises (lower and less variable for Causal-AIRL); environment steps per iteration decline similarly for both; the posterior KL in Causal-AIRL warms up then saturates, indicating a well-behaved latent; policy loss stabilises earlier under Causal-AIRL, consistent with a calmer reward signal. These traces corroborate non-degenerate learning and support the invariance-based mechanism.

Chapter 7

Discussion

This chapter interprets the findings from Chapter 6, makes assumptions explicit, and draws out implications within our stated scope: tabular GridWorlds with fully observed, known dynamics and simulated Z .

Across the confounded and held-out protocols, the pattern is consistent. In Fig. 6.1, Causal-AIRL improves cross- Z policy agreement in both directions. The matched-slice variance summary in Fig. 6.2 supports a mechanism in which gains arise from suppressing Z -specific fluctuations that matter for behaviour rather than wholesale reward re-scaling. Under the explicit generalisation protocol, the direction of the effect persists (Fig. 6.4). Absolute agreement levels vary with topology—corridor geometry and obstacle placement alter policy structure—but the qualitative story is stable.

In clean regimes, Causal-AIRL does not pay a price for its invariance bias. Sweeps over demonstrations, slip, reward type, and γ (Figs. 6.5–6.7) show reward Spearman parity within confidence intervals, with secondary metrics moving in tandem. This matches the objective’s design: when Z has no explanatory power, the variance penalty is effectively idle and the method behaves like AIRL with minor regularisation. Dense shaping compresses gaps by construction, whereas sparse rewards magnify rank-level differences; the sweeps make this clear. Practically, where data are clean and fully observed, adopting Causal-AIRL neither harms reward fidelity nor degrades policy quality, so the invariance terms act as a robustness safety net that does not exact a toll in benign settings.

We now turn to cost and stability, and explain the reading of the scaling curves. We deliberately use small grids for Fig. 6.8 to run many short, statistically independent experiments with tight intervals; we trade depth for breadth to prioritise statistical validity. This also yields a fair comparison for classical baselines: Ng–Russell and

MaxEnt are particularly strong in small tabular MDPs where exact dynamic programming or convex optimisation is tractable. The resulting picture is that, on problems satisfying tabular assumptions with modest state spaces, classical IRL is competitive, AIRL provides a robust deep comparator, and Causal-AIRL shifts the accuracy–time frontier slightly while preserving comparable overall cost trends. Causal-AIRL adds mechanism-specific overhead from latent sampling (default $K=5$) and extra discriminator terms; this overhead is largely orthogonal to state-space size. Extrapolation to much larger grids should therefore be viewed as a representation-learning question rather than a tabular extension. Hardware, wall-clock protocol, and grid configurations appear in D and A.

Two constraints anchor the above. First, the setting is fully observed with known dynamics and simulated Z , isolating the reward-learning effect but limiting realism; partial observability and unknown Z would require learning a proxy and addressing reward identifiability. Second, our across- Z variance aggregate weights all (s, a) equally; low-visitation states can influence the summary. Agreement rollouts use identical seeds to reduce evaluation noise; conclusions rely on mean shifts rather than isolated outliers.

For practitioners, the recommendations are simple. When demonstrations plausibly encode multiple styles or context—or when deployment requires crossing domains—Causal-AIRL is preferred because it improves transfer across Z while preserving clean-regime fidelity and adding only a modest, tunable compute overhead via K . In single-expert, dense-reward, fully observed tasks with tight budgets, AIRL remains a strong baseline; adopting Causal-AIRL does not harm performance in these regimes and offers a safety margin under mild shift. Implementation keys and configuration ranges are listed in A and B.

The most credible next steps follow directly. Extending to partially observed settings with learned proxies for Z raises identifiability questions for reward-level invariance. Moving beyond grids will require surrogate visualisations and bin-free metrics suitable for continuous state spaces. On metrics, adding support-masked invariance, worst- Z summaries, and compact train/test tables aggregated across maps would sharpen the invariance story. Finally, comparing Causal-AIRL to a non-adversarial causal-IRL baseline under identical protocols would separate the contribution of the adversarial game from the invariance term.

Across method-wise aggregates, mean reward Spearman pooled over all configurations is higher for AIRL (Table E.17). Transfer-focused evaluations tell a complementary story: cross- Z policy agreement (Table E.1) and behaviour in held-out regions

(Tables E.3–E.7) consistently favour Causal-AIRL at the policy level. This divergence is expected: the invariance term targets robustness under latent shifts, so improvements are most visible when training and evaluation contexts differ. Because rank statistics are invariant to monotone shaping and value correlations can flip sign under potential terms, policy agreement is the most diagnostic signal under confounding.

Taken together, conditioning on Z and penalising per- (s, a) reward variance yields more transferable policies under confounding, maintains parity in clean regimes, and does so with a controllable and modest compute cost. The assumptions and their consequences are explicit; the limitations suggest tractable extensions rather than contradictions; and the practical guidance aligns with what the curves show. Framed this way, the contribution of this chapter is to synthesise how invariance helps produce policy-useful rewards under latent shift, delineate the limits of our controlled setting, and chart concrete directions for extending robust reward recovery beyond tabular grids.

Chapter 8

Conclusions

This dissertation examined robust reward recovery in inverse reinforcement learning under latent confounding. We evaluated classical IRL (Ng–Russell; MaxEnt), AIRL, and Causal-AIRL—an AIRL variant that conditions on a latent Z and penalises per- (s, a) reward variance across Z —in fully observed tabular GridWorlds with known dynamics. The primary metric was reward Spearman, complemented by value correlation and policy agreement; invariance was summarised by the across- Z reward variance. The study covered clean scenario sweeps, confounded experts, generalisation to held-out regions, and compute–accuracy scaling.

Empirically, Causal-AIRL improved cross- Z transfer in both directions under confounding (Fig. 6.1) and exhibited smoother behaviour in masked regions (Fig. 6.3), consistent with a reward less sensitive to latent context. A matched-scenario variance slice (Fig. 6.2) is consistent with gains arising from suppressing Z -specific fluctuations at salient (s, a) rather than from wholesale rescaling of rewards. In clean regimes, Causal-AIRL exhibited reward Spearman parity with AIRL within confidence intervals, with value and policy metrics moving in tandem across demonstrations, slip, reward type, and γ (Figs. 6.5–6.7), indicating no performance penalty when Z -structure is irrelevant. Scaling placed Causal-AIRL near AIRL on the accuracy–time frontier (Fig. 6.8), with a modest, mechanism-specific overhead attributable to latent sampling (K) and extra discriminator terms; optimisation was stable across runs (Fig. 6.9).

Method-wise mean Spearman aggregated over all runs is higher for AIRL (Table E.17), whereas cross- Z transfer and held-out behaviour favour Causal-AIRL (Tables E.1, E.3–E.7). This divergence reflects the design goal: the invariance bias primarily helps when evaluation departs from the training latent context.

Several assumptions were deliberate. Full observability and known dynamics iso-

late reward-learning effects and provide a strong point of comparison with classical baselines. The latent Z was simulated and unobserved to the learner, enabling controlled tests of invariance and cross- Z transfer. The across- Z variance aggregated uniformly over (s, a) , a conservative summary that may overweight low-visitation states. Scaling experiments used small grids to favour statistical reliability over a few long runs and to benchmark fairly where classical methods are strongest; extrapolation to 100×100 or larger should be framed as a representation-learning problem (feature capacity, locality, and sample complexity) rather than a direct tabular extension. Agreement rollouts use identical seeds to reduce evaluation noise, which can compress apparent variability; conclusions are based on mean shifts rather than isolated outliers. Hardware and timing details appear in D.

For practice, the guidance is clear. When demonstrations plausibly mix operator styles or deployment must cross contexts, Causal-AIRL is preferred: it improves cross- Z agreement and behaviour in held-out regions while maintaining reward fidelity in benign settings, and its compute overhead is tunable via the latent-sample count K . In single-expert, dense-reward tasks with tight budgets, AIRL remains a strong baseline; adopting Causal-AIRL provides a safety margin under shift without harming clean performance. Classical IRL remains competitive on small tabular problems and anchors evaluation.

Natural extensions follow from observed sensitivities. First, move beyond full observability by learning proxies for Z in partially observed settings and studying when reward-level invariance remains identifiable. Second, develop bin-free visualisations and metrics for continuous state spaces to avoid discretisation artefacts. Third, refine robustness metrics with support-masked invariance and worst- Z summaries alongside uniform aggregates. Finally, compare Causal-AIRL to a non-adversarial causal-IRL baseline under identical protocols to isolate the role of the adversarial game.

In sum, conditioning on Z and penalising per- (s, a) reward variance yields rewards that transfer better under confounding while preserving parity in clean regimes at modest cost. Read alongside the controlled benchmark and the invariance-oriented metrics introduced here, these results support invariance-regularised reward learning as a practical default when robustness to latent variation matters, and they outline a credible path toward settings where confounders are learned rather than given.

Bibliography

- [1] Pieter Abbeel and Andrew Y Ng. Apprenticeship learning via inverse reinforcement learning. In *Proceedings of the twenty-first international conference on Machine learning*, page 1, 2004.
- [2] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.
- [3] Ioana Bica, Daniel Jarrett, and Mihaela van der Schaar. Invariant causal imitation learning for generalizable policies. *Advances in Neural Information Processing Systems*, 34:3952–3964, 2021.
- [4] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.
- [5] Pim De Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. *Advances in neural information processing systems*, 32, 2019.
- [6] Chelsea Finn, Sergey Levine, and Pieter Abbeel. Guided cost learning: Deep inverse optimal control via policy optimization. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 49–58. PMLR, 2016.
- [7] Justin Fu, Katie Luo, and Sergey Levine. Learning robust rewards with adversarial inverse reinforcement learning. *arXiv preprint arXiv:1710.11248*, 2017.
- [8] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario March, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of machine learning research*, 17(59):1–35, 2016.

- [9] Seyed Kamyar Seyed Ghasemipour, Richard Zemel, and Shixiang Gu. A divergence minimization perspective on imitation learning methods. In *Conference on robot learning*, pages 1259–1277. PMLR, 2020.
- [10] Dylan Hadfield-Menell, Stuart J Russell, Pieter Abbeel, and Anca Dragan. Co-operative inverse reinforcement learning. *Advances in neural information processing systems*, 29, 2016.
- [11] Jonathan Ho and Stefano Ermon. Generative adversarial imitation learning. *Advances in neural information processing systems*, 29, 2016.
- [12] Ilya Kostrikov, Ofir Nachum, and Jonathan Tompson. Imitation learning via off-policy distribution matching. *arXiv preprint arXiv:1912.05032*, 2019.
- [13] David Krueger, Ethan Caballero, Joern-Henrik Jacobsen, Amy Zhang, Jonathan Binas, Dinghui Zhang, Remi Le Priol, and Aaron Courville. Out-of-distribution generalization via risk extrapolation (rex). In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 5815–5826. PMLR, 18–24 Jul 2021.
- [14] Jan Leike, David Krueger, Tom Everitt, Miljan Martic, Vishal Maini, and Shane Legg. Scalable agent alignment via reward modeling: a research direction. *arXiv preprint arXiv:1811.07871*, 2018.
- [15] Sergey Levine and Vladlen Koltun. Guided policy search. In *International conference on machine learning*, pages 1–9. PMLR, 2013.
- [16] Andrew Y Ng, Stuart Russell, et al. Algorithms for inverse reinforcement learning. In *Icml*, volume 1, page 2, 2000.
- [17] Takayuki Osa, Joni Pajarinen, Gerhard Neumann, J Andrew Bagnell, Pieter Abbeel, Jan Peters, et al. An algorithmic perspective on imitation learning. *Foundations and Trends® in Robotics*, 7(1-2):1–179, 2018.
- [18] Judea Pearl. *Causality*. Cambridge university press, 2009.
- [19] Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78(5):947–1012, 2016.

- [20] Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of causal inference: foundations and learning algorithms*. The MIT press, 2017.
- [21] Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, Inc., USA, 1st edition, 1994.
- [22] Nathan Ratliff, J Andrew Bagnell, and Martin Zinkevich. Maximum margin planning. In *Proceedings of the 23rd international conference on Machine learning*, pages 729–736. ACM, 2006.
- [23] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pages 627–635. JMLR Workshop and Conference Proceedings, 2011.
- [24] Kangrui Ruan, Junzhe Zhang, Xuan Di, and Elias Bareinboim. Causal imitation learning via inverse reinforcement learning. In *The Eleventh International Conference on Learning Representations*, 2023.
- [25] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.
- [26] Richard S Sutton, Andrew G Barto, et al. *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge, 1998.
- [27] Umar Syed and Robert E Schapire. A game-theoretic approach to apprenticeship learning. *Advances in neural information processing systems*, 20, 2007.
- [28] Markus Wulfmeier, Peter Ondruska, and Ingmar Posner. Maximum entropy deep inverse reinforcement learning. *arXiv preprint arXiv:1507.04888*, 2015.
- [29] Brian D Ziebart. *Modeling purposeful adaptive behavior with the principle of maximum causal entropy*. Carnegie Mellon University, 2010.
- [30] Brian D Ziebart, Andrew L Maas, J Andrew Bagnell, Anind K Dey, et al. Maximum entropy inverse reinforcement learning. In *Aaai*, volume 8, pages 1433–1438. Chicago, IL, USA, 2008.

Appendix A

Reproducibility Details

This appendix consolidates the minimal details necessary to reproduce the experimental results reported in Chapters 5–6. File paths are given in `monospace`, limited to the top-level source archive, so that interested readers may locate exact definitions.

Environment definitions

GridWorld environments are defined in `envs/environments.py`. Layouts specify grid sizes, goal and terminal states, and any obstacle placements; actions are the four cardinal moves $\{\uparrow, \downarrow, \leftarrow, \rightarrow\}$, with slip probability p replacing the intended move uniformly at random with one of the other three. The discount factor γ and transition kernel T are set explicitly in the environment constructors. These definitions yield the tabular MDPs used in all sweeps (cf. Chapter 5.1).

Ground-truth rewards

Ground-truth reward maps are constructed in `experiments/eval.py`. Two designs are provided: sparse rewards, which assign non-zero values only at designated terminal states; and shaped rewards, which add a potential-based term to provide dense guidance. The shaping potentials are explicitly coded to preserve policy invariance under reward transformation, ensuring that shaped and sparse regimes bracket typical IRL settings.

For the MaxEnt baseline, the planner temperature τ defaults to 1.0 unless overridden; this matches `configs/gridworld_baseline_maxent.yaml`. Scenarios that deviate from this explicitly set τ in their config files.

Expert generation

Expert demonstrations are generated by an entropy-regularised planner implemented in `envs/environments.py`. A soft-optimal policy is computed with a tunable temperature parameter, yielding stochastic but near-optimal trajectories. Clean sweeps use $N \in \{5, 10, 20, 50\}$; the confounded suite uses $N \in \{10, 20, 40\}$. Values $N \in \{1, 2\}$ were used only for sanity checks and not reported in the main figures.

Training configuration keys

All hyperparameters are passed through YAML configuration files under `configs/`. Names are aligned with Table 4.1 (*training knobs*) in the main text. Keys include:

- `entropy_coef`, `grad_clip_norm`, `reward_logit_clip`
- `kl_coeff`, `kl_warmup_epochs`, `inv_coeff`
- `latent_dim`, `num_z_samples`

Baselines such as `gridworld_baseline_airl.yaml` and `gridworld_baseline_causal_airl.yaml` pin the settings used in Figures 6.1–6.9.

Optimisers

The policy update is implemented in `irl/airl.py` (and inherited in `irl/causal_airl.py`) using REINFORCE with a learned baseline, advantage normalisation, entropy regularisation, and gradient clipping. The discriminator/critic is trained with Adam; learning rate and (β_1, β_2) values are set via configs. Causal-AIRL extends the discriminator objective with KL and invariance terms (Eq. 4.8), but otherwise shares the optimiser recipe.

Evaluation protocol

Metrics are defined in `experiments/eval.py`. Primary fidelity is Spearman rank correlation between learned and ground-truth rewards. Secondary metrics are value correlation (Pearson between induced V^π) and policy agreement (fraction of matched actions on seeded rollouts). Reward variance across latent Z is computed using $K = 5$

Monte Carlo samples per (s, a) , averaged uniformly. All metrics are clipped to $[-1, 1]$ for numerical stability. Confidence intervals (95%) are obtained by bootstrap with 1,000 resamples.

Figure regeneration

Plots are generated from scripts under `visualisation/`. Each script reads results in `results/` directories and reproduces overlays, sweeps, and scaling curves. Config templates in `configs/` provide the exact settings.

Appendix B

Hyper-parameters

Final hyperparameters were selected via the scripts in `scripts/` and fixed for all reported runs. Best values are:

- AIRL: `entropy_coef = 0.005, grad_clip_norm = 1.0`
- Causal-AIRL: `kl_coeff = 0.001, inv_coeff = 0.02, latent_dim = 8`

Hyperparameter sweeps are scripted in:

- `scripts/run_airl_hparam_sweep.sh,`
`scripts/run_airl_scenario_sweep.sh`
- `scripts/run_causal_airl_hparam_sweep.sh,`
`scripts/run_causal_airl_scenario_sweep.sh`

These sweep over ranges of entropy coefficients, gradient clips, KL warm-up epochs, and invariance weights. Training schedules were fixed by family. Baseline and sweep runs (hyper-parameter and scenario sweeps) used a maximum of 100 epochs. Confounded, generalisation, and scaling runs used a maximum of 200 epochs. We did not employ patience or early stopping; all runs trained to the stated epoch budget.

Appendix C

Notation and Symbols

The notation summary appears in 3.1. For completeness, extended forms include:

- $r_{\theta}(s, a, z)$, the Causal-AIRL reward conditioned on latent z ;
- $\text{Var}_z(r_{\theta}(s, a, z))$, the per- (s, a) invariance penalty (Eq. 4.7);
- L_{KL} and L_{inv} , auxiliary discriminator terms (Eqs. 4.6–4.7);
- K , the number of Monte Carlo samples for invariance estimation.

Equation variants omitted from the main text include the explicit unbiased variance estimator used in implementation, which appears in `irl/causal_irl.py` as a small-sample correction.

Appendix D

Hardware, Software, and Scaling

Hardware

All experiments were run locally on a laptop with the specifications in Table D.1.

Component	Specification
CPU	Intel® Core™ i7-8750H @ 2.20 GHz (6 cores / 12 threads)
RAM	12.7 GB
GPU (discrete)	NVIDIA GeForce GTX 1050
GPU (integrated)	Intel UHD Graphics 630 (unused for training)
Storage	SSD

Table D.1: Hardware specifications of the experimental platform.

Software

- Operating system: Windows 11 (Version 20H2, Build 22H2).
- Python 3.10 (project requires ≥ 3.10 , < 3.12).
- Core libraries: `numpy`, `scipy`, `pandas`, `torch`, `matplotlib`, `PyYAML`, `networkx`, `psutil`, `tabulate`, and `gymnasium[classic_control]`.
- Development/testing: `pytest` (not used at runtime).
- Dependency management: `setuptools` with `pyproject.toml` and `setup.py`; pinned list in `requirements.txt` and `environment.yaml`.

Wall-time measurement protocol

Wall-clock times for scaling experiments (Fig. 6.8) were measured using `run_scaling_performance.sh`, which wraps calls to `experiments/run_experiment.py` with UNIX timestamp logging:

```
START=$(date +%s)
... run experiment ...
END=$(date +%s)
echo $((END-START)) >> results/scaling/perf.csv
```

Overrides match the scaling script: *env.reward_type = sparse*, *env.slip_prob = 0.0*, and *irl.max_iters = 200*; grid sizes are $N \in \{5, 7, 9, 11, 13, 15\}$ with five seeds. Each configuration was repeated with five seeds across grid sizes $N \in \{5, 7, 9, 11, 13, 15\}$. Hyperparameters were fixed (AIRL entropy, clip; Causal-AIRL KL, invariance, latent dimension) and injected via `-override` flags. Logs are stored under `results/logs/scaling_performance.log`. Curves report means with 95% bootstrap intervals over 1,000 resamples.

Appendix E

Aggregated Tables & CI Summaries

This appendix collects bootstrap summaries that back up Chapter 6. Unless stated otherwise, all confidence intervals are nonparametric bootstrap with $B=1000$ resamples and 95% confidence. Columns labelled “mean [low, high]” are a shorthand for the point estimate and the associated interval. Where shown, $d_{\text{vs base}}$ is a standardised mean difference (Cohen’s d) versus the stated baseline; positive values favour the row relative to the baseline.

Unless noted, “ Δ vs baseline” reports a standardised mean difference relative to the stated baseline. For a metric x , with method-group mean $\bar{x}_m$, baseline mean $\bar{x}_b$, sample sizes n_m, n_b , and sample standard deviations s_m, s_b , we compute

$$d = \frac{\bar{x}_m - \bar{x}_b}{s_p}, \quad s_p = \sqrt{\frac{(n_m - 1)s_m^2 + (n_b - 1)s_b^2}{n_m + n_b - 2}}.$$

Confidence intervals shown in the tables are constructed on group means via bootstrap ($B=1000$, 95%). The standardised difference d is reported without a bootstrap interval for compactness and interpretability on a common scale.

E.1 Cross-Z Transfer: Policy Agreement by Method and Demonstrations

This table summarises cross-Z policy agreement under confounding, grouped by `method × env.reward_type × expert.num_trajectories`. Higher agreement is better. Baseline is (AIRL, sparse, $N=10$).

Table E.1: Cross-Z transfer: policy agreement (higher is better). Bootstrap $B=1000$, 95% CIs. Grouped by method, reward type, and demonstrations N . Baseline: AIRL, sparse, $N=10$.

Method	Reward	N	n	Mean	95% CI	Δ vs baseline
AIRL	sparse	10	30	0.237	[0.187, 0.289]	0.000
AIRL	sparse	20	90	0.257	[0.234, 0.281]	0.165
AIRL	sparse	40	30	0.217	[0.184, 0.253]	-0.165
Causal-AIRL	sparse	10	30	0.380	[0.337, 0.424]	1.064
Causal-AIRL	sparse	20	225	0.373	[0.355, 0.389]	1.021
Causal-AIRL	sparse	40	30	0.375	[0.321, 0.425]	0.937

E.2 Held-out Generalisation: Confounded + Spatial Masks

We report three metrics in held-out regions (top-left, top-right, bottom-left, bottom-right): reward Spearman (rank fidelity; higher is better), value correlation (scale-sensitive; sign can be influenced by potentials), and policy agreement (higher is better). Baseline is (bottom_left, AIRL).

Reward Spearman (held-out regions)**Table E.3:** Held-out regions: reward Spearman (higher is better). Bootstrap $B=1000$, 95% CIs. Baseline: bottom left, AIRL.

Region	Method	n	Mean	95% CI	Δ vs baseline
bottom left	AIRL	15	0.179	[0.160, 0.196]	0.000
bottom left	Causal-AIRL	15	-0.074	[-0.151, 0.011]	-2.110
bottom right	AIRL	15	0.202	[0.185, 0.219]	0.631
bottom right	Causal-AIRL	15	-0.040	[-0.096, 0.025]	-2.440
top left	AIRL	15	0.224	[0.212, 0.239]	1.342
top left	Causal-AIRL	15	-0.011	[-0.089, 0.066]	-1.584
top right	AIRL	15	0.199	[0.180, 0.219]	0.513
top right	Causal-AIRL	15	-0.002	[-0.092, 0.096]	-1.223

Value correlation (held-out regions)**Table E.5:** Held-out regions: value correlation (higher is better). Bootstrap $B=1000$, 95% CIs. Baseline: bottom left, AIRL.

Region	Method	n	Mean	95% CI	Δ vs baseline
bottom left	AIRL	15	-0.409	[-0.547, -0.173]	0.000
bottom left	Causal-AIRL	15	0.073	[-0.303, 0.444]	0.790
bottom right	AIRL	15	-0.769	[-0.913, -0.615]	-0.984
bottom right	Causal-AIRL	15	-0.288	[-0.428, -0.160]	0.353
top left	AIRL	15	0.842	[0.613, 0.999]	3.115
top left	Causal-AIRL	15	0.468	[0.234, 0.692]	2.046
top right	AIRL	15	-0.107	[-0.390, 0.200]	0.602
top right	Causal-AIRL	15	-0.359	[-0.669, 0.018]	0.087

Policy agreement (held-out regions)

Table E.7: Held-out regions: policy agreement (higher is better). Bootstrap $B=1000$, 95% CIs. Baseline: bottom left, AIRL.

Region	Method	n	Mean	95% CI	Δ vs baseline
bottom left	AIRL	15	0.111	[0.011, 0.256]	0.000
bottom left	Causal-AIRL	15	0.344	[0.222, 0.478]	0.897
bottom right	AIRL	15	0.025	[0.000, 0.075]	-0.432
bottom right	Causal-AIRL	15	0.058	[0.017, 0.117]	-0.262
top left	AIRL	15	0.583	[0.400, 0.733]	1.600
top left	Causal-AIRL	15	0.333	[0.233, 0.450]	0.905
top right	AIRL	15	0.544	[0.433, 0.633]	1.834
top right	Causal-AIRL	15	0.222	[0.089, 0.367]	0.403

E.3 Scaling Curves: Accuracy, Time, and Sample Use

We aggregate across grid sizes to show mean accuracy and cost by method. Absolute wall times are hardware-specific; trends are most informative. Baseline for $d_{\text{vs base}}$ is AIRL.

Reward Spearman (scaling)

Table E.9: Scaling: reward Spearman (higher is better). Aggregated across grid sizes; bootstrap 95% CIs. Baseline: AIRL.

Method	n	Mean	95% CI	Δ vs baseline
AIRL	590	0.223	[0.212, 0.234]	0.000
Causal-AIRL	1040	0.103	[0.091, 0.117]	-0.650
MaxEnt	35	-0.147	[-0.174, -0.122]	-2.693
Ng–Russell	35	0.244	[0.235, 0.253]	0.157

Value correlation (scaling)**Table E.11:** Scaling: value correlation (higher is better). Baseline: AIRL.

Method	<i>n</i>	Mean	95% CI	Δ vs baseline
AIRL	590	0.087	[0.045, 0.128]	0.000
Causal-AIRL	1040	-0.201	[-0.216, -0.185]	-0.774
MaxEnt	35	0.355	[0.279, 0.428]	0.540
Ng–Russell	35	-0.075	[-0.231, 0.084]	-0.321

Policy agreement (scaling)**Table E.13:** Scaling: policy agreement (higher is better). Baseline: AIRL.

Method	<i>n</i>	Mean	95% CI	Δ vs baseline
AIRL	590	0.502	[0.484, 0.522]	0.000
Causal-AIRL	1040	0.429	[0.417, 0.442]	-0.338
MaxEnt	35	0.780	[0.754, 0.804]	1.243
Ng–Russell	35	0.420	[0.347, 0.495]	-0.360

Wall-clock time and environment steps (scaling)

Table E.15: Scaling: wall-clock time (seconds) and environment steps (lower is better). Base-line: AIRL.

Wall-clock time (s)			
Method	n	Mean	95% CI
AIRL	590	98.430	[93.669, 103.854]
Causal-AIRL	1040	144.473	[139.519, 150.102]
Environment steps			
Method	n	Mean	95% CI
AIRL	590	985.997	[956.865, 1020.000]
Causal-AIRL	1040	1012.287	[994.928, 1030.819]

E.4 Method-wise Summary (All Runs)

Aggregates across all configurations by method. Reward-variance across Z is only defined for methods producing $r(s, a, z)$; entries are shown as em-dash where not applicable.

Table E.17: Summary by method: reward Spearman (higher is better).

Method	n	Mean	95% CI
AIRL	590	0.223	[0.211, 0.234]
Causal-AIRL	1040	0.103	[0.091, 0.115]
MaxEnt	35	-0.147	[-0.173, -0.122]
Ng–Russell	35	0.244	[0.235, 0.252]

Table E.19: Summary by method: value correlation (higher is better).

Method	<i>n</i>	Mean	95% CI
AIRL	590	0.087	[0.048, 0.130]
Causal-AIRL	1040	-0.201	[-0.216, -0.185]
MaxEnt	35	0.355	[0.278, 0.427]
Ng–Russell	35	-0.075	[-0.228, 0.090]

Table E.21: Summary by method: policy agreement (higher is better).

Method	<i>n</i>	Mean	95% CI
AIRL	590	0.502	[0.484, 0.520]
Causal-AIRL	1040	0.429	[0.417, 0.442]
MaxEnt	35	0.780	[0.754, 0.805]
Ng–Russell	35	0.420	[0.345, 0.490]

Table E.23: Summary by method: across-Z reward variance (lower is better).

Method	<i>n</i>	Mean	95% CI
Causal-AIRL	285	0.117	[0.092, 0.147]

E.5 Reward Type × Method Summary

Aggregates split by reward design. Shaped rewards tend to raise correlations uniformly; sparse rewards stress credit assignment. Small *n* for some classical entries yields wider intervals.

Table E.25: Reward type $\times$ method: Spearman, value correlation, and policy agreement (means with 95% CIs).

Reward	Method	Spearman		Value corr.		Policy agr.	
		Mean	95% CI	Mean	95% CI	Mean	95% CI
shaped	AIRL	0.204	[0.170, 0.236]	0.335	[0.274, 0.396]	0.586	[0.566, 0.607]
shaped	Causal-AIRL	0.326	[0.310, 0.341]	-0.143	[-0.161, -0.126]	0.451	[0.431, 0.472]
sparse	AIRL	0.242	[0.237, 0.246]	-0.019	[-0.068, 0.037]	0.454	[0.427, 0.478]
sparse	Causal-AIRL	-0.020	[-0.028, -0.012]	-0.252	[-0.274, -0.231]	0.408	[0.392, 0.422]
sparse	MaxEnt	-0.266	[-0.283, -0.243]	0.089	[0.058, 0.108]	0.744	[0.656, 0.840]
sparse	Ng–Russell	0.263	[0.257, 0.266]	-0.675	[-0.688, -0.651]	0.096	[0.080, 0.128]

Caveats. (i) Some entries have small n (e.g., classical baselines in large grids), widening intervals; (ii) value correlation can be negative under potential shaping despite reasonable policy agreement; (iii) across- Z variance is a uniform (s, a) -average and complements support-masked variants reported in Appendix B (see also Section 5.3).